Light Semantic Harness: Applied Study

Blame Repository Engineering Study

October 27, 2026

Abstract

Prioritize audited success and truthful evidence before resources at comparable quality. Current Light combines syntax navigation, configured JS/TS identity and guarded edits behind five primitives. Historical 21/36 versus integrated Cline 28/36 and incomplete consumption do not describe current code. F57 adds ordinary-search declaration routes, observed body-edit headers and failed-check priority while preserving final-source obligations. F59–F61 execute 54 registered assignments with two repetitions: isolated known-task repair/header panels and fresh cross-module TS-Pattern requests. F59 saves tokens without observed direct repair uptake; F60 retains two functional failures despite passing authored tests, with no header exposure in those trials. F61 audits reference/current/Cline at 5/6, 6/6, 6/6; the reference failure is a new test's compilation, not production behavior. That bundle uses 54.2% fewer tokens and 38.6% lower SDK estimated cost than Cline, with 27.4% longer agent time. Token savings survive task/pair omission; the registered duration/resource gate fails. All 54 original patches reproduce their bounded endpoint outcomes. These small panels establish neither causal mediation nor general efficiency. F62 adds bounded validation history, boundary sensitivity controls and passive HTTP timing. Its eighteen known-task assignments audit reference/candidate/Cline at 4/6, 5/6, 6/6; no candidate receives the new invalidation feedback, so no model benefit is established. Lower candidate tokens coexist with lower overall quality than Cline and 8.9% longer time. Cache/invoice limits and unknown F52 consumption remain explicit; Z3 stays optional and is disabled in the new panels. F65 resolves a formatter-observer blind spot and demonstrates actual SDK clipping with loopback controls: exact scope metadata survives in 4/4 corrected cases versus 2/4 raw cases. F66 finds no batch use in four trials and rejects that prototype. F67 observes failed-compiler-first order in 4/4 corrected runs versus 0/4 reference, at equal bounded correctness but 4.6% more tokens. This is transport/repair reliability, not a general efficiency result. Finite experiments cannot establish exhausted improvement margin.

1 Scope, versions and evidence boundaries

This compact report separates implemented behavior, historical measurements, new measurements and proposals. Expanded phase-by-phase notes are preserved verbatim, with byte/hash inventory, in `docs/experiments/light-harness-study-history-2026-10-15`. Original campaign artifacts remain separate. Neither dependent repeats nor local replay cases count as independent real-issue confirmations.

F28–F31, F34–F36, F38b, F41–F44, F47 and F50 use `openai-codex/gpt-5.6-luna/high` and Cline SDK 0.0.82. Native comparison arms include a pinned research search repair; these are not unmodified CLI comparisons. Actual schemas, prompt sources, model/effort, source/checker/environment hashes, ordered assignments and budgets are retained. Light emits seven tools in these runs: five harness primitives and optional skills/web support; native emits five. Normalized contracts are not literally identical dynamic prompts. Checkout labels and review are unblinded; provider revision/seeds and complete cache/invoice accounting remain unknown.

Four claims remain distinct: tests executed, tests passed, verification admitted for a source/environment/scope, and satisfaction of user criteria. A process exit or model summary

alone proves neither the latter two nor an exhaustive test run. Unknown counters are not zero; pending, skipped and runner-specific todo labels must not be silently interchanged. Raw recorded output need not be model-visible.

2 Implemented behavior

Primitive	Responsibility and boundary
`search_code`	Syntax-derived declarations and bounded lexical search. Explicit configured-project JS/TS alias or `path##qualifiedName` references use the TypeScript service; bare names, unsupported languages and unavailable projects retain labeled lexical fallback. Discovery grants no edit coverage or runtime-completeness proof. Multiple scopes use arrays; literal spaces are not an implicit union. Bare-name fallback and ordinary searches additionally offer bounded exact syntax declaration candidates requiring explicit choice; same-file homonyms retain distinct aliases. Signatures are capped at 160 code points with an explicit truncation flag and non-exhaustiveness warning. Production retains the bounded lexical OR collection.
`read_code`	Files, ranges, symbols, signatures/outlines, changes and check journals. Small automatic reads can expose a complete file; large files route to outlines. Signature reads omit implementations and cannot authorize unseen body edits. The SDK can execute several read calls within one model iteration. The unused F66 batch prototype was removed after evaluation. Only actually emitted bytes establish consumer/context-specific observation. Truncation provides bounded continuation routes. A confirmed missing literal path can now return at most five exact-basename indexed candidates, with explicit next reads. It never selects a target or grants edit coverage; large ambiguous sets, protected files and symlinks are excluded.
`edit_code`	Symbol operations and explicit file patches. Whole-symbol replacement and body-only replacement differ; the latter preserves parameters. Configured compiler rename preserves identity/aliases and remains bounded. Symbol writes require observed coverage; file patches require exact preimages and explicit scope. Stale/ambiguous targets and invalid batches are refused. Body-edit receipts show bounded observed input headers, including first noops and rejected syntax. Repeated noops additionally offer a complete read route; other explicit batch operations may change the header or imports. Commit locks, durable pending records and recovery preserve conflicts; they do not isolate all external editors/processes.
`run_task`	Project tasks, literal commands, dependencies and running handles. Capture process outcome, source effects and bounded diagnostics. The default initial wait is bounded at 15 seconds; handles/cursors support continuation and cancellation. Project test arguments remain literal/bounded. A successful transport is distinct from command or test success. Exact failed project invocations produce advisory repair priority, without running commands. Recognized direct literal no-emit TypeScript invocations now retain invocation-only classification; an exact failed command can lead repair without implied file coverage.

Primitive	Responsibility and boundary
`finish_task`	Reuse/refresh relevant attempted checks and explicit required selections, prioritize known failed selections, stop on refreshed code failures, and report delivery separately from admitted verification. After source repair, a recognized failed literal no-emit compiler can refresh first; original required checks still follow. Unverified delivery can remain possible with reservations. Arbitrary user behavior and truthful prose are not independently established by this primitive.

`LightWorkspace`, `light-tools.ts`, `light-contracts.ts`, `light-aliases.ts`, `light-edit.ts`, `light-finish-service.ts` and the executor port own these responsibilities under `packages/agent-runtime/src/harness`. Syntax adapters use ast-grep/Tree-sitter. Configured compiler identity is more precise than lexical references, but partial projects, missing imports and unsupported languages retain explicit limits. This combines compact tools with project-aware navigation; no controlled Serena performance comparison is claimed. Historical harness-core contracts are not automatically current Light behavior.

`workspace-command-process.ts` bounds capture and abnormal process-group cleanup; handles are session-local, not restart-persistent. Command policy uses actual managed/isolation metadata and is not a universal shell sandbox. Current check journals retain observed counts and source identity while formal admission may remain unverified. Stable provider projection preserves an unchanged canonical prefix; compaction resets its basis. Context bytes and cache hits are not interchangeable with billed tokens. An opt-in `run_task` analysis operation translates a restricted, explicitly typed TypeScript scalar function and supplied property into Z3 constraints. Its schema is hidden by default and its engine loads on demand. Integer/Boolean domains, safe intermediate arithmetic and syntax/path/process/output limits are explicit. Reproduced witnesses, bounded absence and unknown outcomes are distinct; source changes invalidate results. No solver result admits source verification, edit coverage, a required invocation or user acceptance. Calls, loops, mutation, globals and general floating-point semantics remain unsupported, except explicitly modeled standard integer `Math.min/max/abs`. Shadowing, visible mutation and escaping Math references are refused; imported side effects remain an explicit assumption. Optional complete-body read hints screen supported syntax without a solver call, inferred property, automatic bounds or edit admission. `BLAME_AGENT_SYMBOLIC_HINTS=false` disables these hints independently of analysis-schema exposure. Invocation-owned reports for supported Vitest 3.2.4 runs behind Node/npm wrappers now retain actual case counts, file hashes, invocation identities and partial environment identities. A masked failed child is failed; an empty selection or unsupported reporter stays unknown. Runtime selection does not establish delivered-test type selection or admitted verification. The local provider-compatible HTTP adapter now restores an explicitly configured output limit on the exact HTTPS Codex Responses endpoint. It preserves stricter limits and cancellation, rejects uninspectable capped requests and adds no default limit. Remote/hub HTTP owners need their own adapter configuration.

3 Historical measurements: keep cohorts separate

The principal historical campaign reports Blame 21/36 versus integrated Cline 28/36. Global consumption is incomplete. Smaller pilots sometimes reduce tokens at lower quality; they do not establish an unconditional efficiency advantage. Their versions predate multiple current fixes.

Later F27 fresh RxJS migration repeats reach 3/3 audited successes per arm. The Light variant measured there uses 19.5% fewer tokens than rename-disabled Light and 59.3% fewer than repaired native on that one migration cluster. Its separate fresh behavioral panel has 4/6 full audited successes per arm, with different failure types. Known Zod diagnostics, missing

responses and adverse repeats remain separate. These are neither independent confirmations per repeat nor F28/F29 data.

Compiler references/rename, POSIX missing-tool recovery and TypeScript member boundaries already exist before F28's reference. In particular, exit-127 missing-executable evidence can permit reserved unverified delivery; it does not turn a source/test failure into a pass. Six later guidance contracts have local controls. Historical defects already repaired in these paths must not be counted as current failures.

4 Diagnosis against current code

Observation	Evidence, attribution and current status
Scope versus absence	F28 date-fns traces use literal compound scopes or `*` and later recover with recursive scans. Root-only `*` can have nonzero candidates while missing nested files. Empty-scope advice exists; nonempty root-only scope is still not a project-wide search. This combines model choice with contract/presentation friction.
Unrelated reads	F28 repetition-two reporting schedules four unrelated helper reads in iteration one, before the target response. The target is only 1,081 characters. The adapter does not automatically expand to these peers; model planning/common instructions, not missing compiler identity, explain the recorded calls. Causal prompt influence remains unisolated.
Late lexical targets	OR terms fill a 32-occurrence pool before ranking. F28 `function keys/values` queries miss late declarations; F29's identical initial `function forIn` query returns line 9 on reference and line 13015 on candidate in all three repeats. Production retains the old ordering after the candidate's adverse quality result.
Edit boundaries	F28's first successful Light Lodash edit contains an options-dependent body replacement in 6/8, 4/8 and 7/8 repeats: 17/24 total. Parameters are repaired later. Scoped imports/multiple headers correctly fail atomically. Whole-symbol replacement already exists. Duplicate JSDoc and indentation show operation-boundary friction; no automatic signature invention, comment stripping or scope widening is justified.
Validation and visibility	Native date-fns records 28,276 characters, but a pinned 8,000-character projection replay loses the aggregate. Qualified unknown counts are accepted. The unchanged RxJS 3ms/8ms window test inconsistently fails on an identical migration endpoint; retain the registered independent failure. Unchanged input passes 6/6 diagnostic invocations and the endpoint 5/6: pristine failure and treatment attribution are not established.
Processes/environment	Inode exhaustion interrupts preparation 14 before any model dispatch. Compact 1,367 historical Git object stores with identities/diffs preserved and resume the original slot; retain the 454.758s administrative gap. A later USD 0.04 soft stop leaves correct code/checks but no delivery and one unfinished read-only Git call. No outcome or invocation is invented.

Observation	Evidence, attribution and current status
Reasoning versus checker coverage	F29 frozen tests miss custom `lodash.iteratee` call/exception differences. Source was available; wrong guard simplifications are model preservation mistakes and the grader has a blind spot. Supplementary evidence rejects user behavior without rewriting frozen grades or attributing a population regression to ranking.
Fresh versus stale names	F29 has five padded `path## name` reads. Four exact controls resolve; conditional-expression helper `getSymbols` is not indexed as that declaration. A later production repair distinguishes an index miss from a stale alias and proposes only unique safe corrective reads. It does not normalize or select a target automatically.

Evidence is SHA-bound in each campaign's `source-review.json`, raw archives and diagnostic records. Complete recorded inputs include the unfinished cost-stop call; its execution outcome stays unknown. Three proposed causal axes organized F28's six factors: actionable navigation, explicit edit/rename boundaries, and reliable validation reporting. None guarantees correct reasoning or prose.

5 New measurements: F28, completed

Four new constructed clusters on three familiar projects: RxJS helper migration (8 bindings, 4 files), date-fns helper migration (30 bindings, 11 files), two-file RxJS promise cancellation, and Lodash own-symbol options. Three dependent repeats and nine rotated serial arms fix 108 assignments. Reference `50680158`, six isolated factors, bundle `5f310425` and repaired native are frozen before dispatch. A common hash-pinned projection changes prompt date only, not clocks.

All 108 actual required final-source checks pass. There are 107 official completions, 107 frozen functional successes and 103 full audited completions. Keep the independent timing failure, soft-cost delivery stop and factual reporting failures. Only the newly supplied RxJS async fixture was corrected to awaited callbacks before paid dispatch; upstream UI limitations remain. Positive, unchanged-source and protected-input controls are preserved.

Main arm	Full audited	Observed tokens	Worker seconds
Reference Light	12/12	1,376,202	883.819
Six-change bundle	12/12	1,768,950	1,045.406
Repaired Cline SDK	12/12	1,992,718	809.281

On the same 12 strict pairs, bundle/reference changes are +28.5% tokens, +3.4% estimated cost and +18.3% worker time. Bundle/native changes are −11.2%, −24.3% and +29.2%; reference/native are −30.9%, −26.8% and +9.2%. The bundle has no demonstrated net advantage over reference. Equal small success counts do not establish population equivalence. Individual-factor subsets differ. Patch/reference shows −4.3% tokens on 12 pairs, but its rich recovery response never activates; schema wording and stochastic paths prevent attribution specifically to that response.

One bundle agent's own timing-test failure triggers a refused finish, inspection, an unchanged-source rerun and truthful disclosure of both outcomes. This is within-trial recovery, not researcher retry or rescue of the independently failed score. Another successful trial has 11 forwarded requests and 10 usage reports, including two identical bodies 2.075s apart. Missing consumption remains unknown; reserve USD 0.23 and exclude it from primary resource pairs. New observed estimate is USD 1.40309660; cumulative observations/reserves then total USD 3.50047236.

6 New measurements: F29, completed; prototype not promoted

Two fresh related option-extension tasks on the familiar Lodash monolith, three dependent repeats and three rotated arms fix 18 assignments. The reference is F28's all-six bundle, not its older reference. The candidate changes only two lexical modules: a second bounded contiguous-term pass and phrase priority, retaining OR fallback, path precedence, scope/limits and observation guards. Node fallback prefers within each visited file, not a global exhaustive ranking. Light schemas, prompt sources, edit and validation modules match. Registration `6b53b2eb` precedes dispatch; no model retry, source repair or feedback occurs.

Local controls retain 16 constructed rows, eight known-query replays and eight additional contract tests (candidate 8/8, reference 6/8). These overlap and are not independent tasks. Twenty existing search tests and both runtimes' 21 guidance contracts pass. Warm local ripgrep medians for two multi-term queries rise from 13–14ms to 18–19ms; single-term search stays near 13ms. This is local API timing, not provider latency. Natural contiguous top snippets occur in 25/31 candidate multi-term calls versus 0/36 reference calls; differing trajectories are not independent randomized mediators.

Arm	Frozen endpoint	Behavior audit	Full audit	Tokens
All-six reference	6/6	6/6	4/6	1,023,482
Ranking candidate	6/6	4/6	4/6	651,343
Repaired native	6/6	5/6	3/6	1,165,430

All 18 frozen endpoints pass; this does not establish all required behavior. Original default `forIn/forInRight` invoke a customized iteratee normalizer on falsy non-null inputs, while `forOwn/forOwnRight` short-circuit them. Candidate repeat two loses default behavior; candidate repeat three and native repeat two eagerly normalize in own mode. A post-selected diagnostic evaluates 56 cases on all nine iterator endpoints (504 correlated probes, not tasks). Original sources/scores stay unchanged; no counterexample reaches the models. This exposes three missed endpoints without claiming independent confirmation or a ranking-caused population regression.

Factual failures stay separate: two reference reports infer pending zero from Node's todo zero/unreported pending; native gives one nonexistent file path and one contradictory no-files-changed claim. Ambiguous additional-declaration wording is not independently penalized when required changes are explicit. Equal full counts therefore conceal different behavioral quality.

Only three strict candidate/reference resource pairs qualify: −33.6% tokens, −21.6% estimated cost and −5.9% worker time. This selected successful subset cannot compensate for lower all-attempt behavioral acceptance. *Do not promote the ranking prototype or claim a global efficiency gain.* Its patch/tests remain reviewable; production lexical ordering stays unchanged. Frozen-functional sensitivity may include the demonstrated semantic failures and is not a comparable-user-quality estimate.

7 New measurements: F30, completed; wording prototype isolated

Three fresh constructed API tasks on previously unmeasured Commander.js 13.1.0 span two or three implementation modules plus public declarations. Three dependent repeats and three rotated serial arms fix 27 assignments. Registration `c940d2c8` precedes dispatch. Reference `79bcf87f` includes both later diagnostic repairs; candidate changes five tool-description initializers only. Runtime execution, input schemas and system prompt sources match. Actual initial requests confirm identical other tool fields and workspace-normalized prompts on all nine Light pairs. Description bytes fall 3,897 to 2,108 and full schemas 9,956 to 8,157. Wording emphasis also changes; length is not independently isolated.

Before dispatch, unchanged inputs fail, authored positives pass supplied/private probes, JS/TS checks and all 108 upstream suites/1,206 tests; eight targeted mutants are rejected. Both runtimes pass 21 existing guidance tests. A protected launcher keeps Jest flags out of `process.argv` and removes host `NO_COLOR` in that test process. Preparation failures/corrections remain separate. Original assertions are unchanged; no model retry, source repair or oracle feedback occurs.

Arm	Frozen endpoint	Behavior audit	Full audit	Tokens
Current reference	9/9	9/9	8/9	1,837,919
Concise wording	9/9	8/9	8/9	1,973,335
Repaired native	9/9	9/9	9/9	2,637,926

All 27 actual required final-source checks pass. Candidate repetition one reconstructs an environment value's origin after a public event listener removes its source variable, so diagnostics name the wrong variable. A supplementary post-selected probe assesses the same three cases on all nine environment endpoints; only that candidate fails. The unchanged primary-only upstream control passes, while the authored local positive also fails the added hook cases. Keep original frozen grades and the control's negative result. These correlated probes are not new confirmation tasks or causal evidence about wording.

Reference repetition two has a separate wording-sensitive reporting failure: its final sentence claims no validation categories were unreported despite absent typecheck counters and Node pending remaining null. Equal full counts therefore conceal different behavioral quality. Audits are performed by the research agent with arm labels visible, without an independent reviewer.

Seven strict candidate/reference pairs yield +7.0% tokens, +12.3% estimated cost and +4.8% worker time. Relaxing only the reporting rejection gives eight pairs and still +11.1%, +15.7% and +10.2%. Both registered gates fail: *keep the wording prototype isolated*. The hidden-argument cluster improves 23.9% in tokens on three pairs, while choices worsen 31.1%; retain both. On eight strict reference/native pairs, resources change −30.2% tokens, −29.7% estimated cost and −0.5% time. That successful subset does not erase the all-attempt delivery difference or establish population equivalence.

Trace evidence still shows body-only edits omitting parameters or containing whole headers, range aliases used as symbols, scoped patch/export/indentation preimage mismatches, repeated replacements and real test failures recovered within the original trials. Guards correctly refuse invalid atomic batches. Shorter descriptions do not eliminate these choices. Extra indentation is hygiene/friction, not an invented functional rejection.

A public-source replay reconstructs all 27 endpoints from archived upstream, fixtures and original diffs, matches every scored source hash, reruns frozen UTC and supplied probes, and reproduces supplementary hooks including the failure. It repeats neither the complete upstream suite nor non-UTC panel by default; their original immutable outcomes remain. No model calls occur in these replays.

8 Later implemented local diagnostics, not model-measured gains

Two small production repairs remain separate from every frozen F28/F29 runtime and are included equally in both F30 Light arms, without an isolated effect test. First, repeated `Update File` headers for one file are labeled `multiple_headers_for_symbol`, preserving the refusal and explicit scope choice. Twenty-five focused tests pass; a replay of 24 original calls reproduces the blocked request with unchanged source. Its new test fails on the old reason.

Second, fresh missing selectors say no indexed declaration matches, rather than claiming an observed symbol disappeared. A padded name receives an exact read suggestion only when the indexed name is unique and cannot resolve to a conflicting or stale alias. Names are not silently trimmed, scope is not widened and hints do not grant observation. Four safety tests and 67

existing identity/coverage tests pass; the same four tests have two expected failures on the frozen old runtime. On five recorded padded inputs, four explicit suggested reads resolve; the unindexed helper remains blocked with a text/range route. This is researcher replay evidence, not measured model adoption or another successful task.

Initial local test-fixture assumptions and an overly strong five-hint replay expectation are retained separately. They are setup corrections, not paid-trial repairs or observed harness regressions. Current targeted suites and lint pass.

9 F31: new issue instances and an unexercised edit repair

The preregistered 18 assignments cover YAML 2.8.1 issue 729 and fast-glob 3.3.3 issue 265, three dependent repeats and three serial rotated arms. Exact upstream, runtime, prompt/schema, model/effort, checker, dependency and budget bindings are archived. Optional anchor-token ranges with clone isolation and synchronous entry filtering with pruning/error propagation use researcher-specified API/edge criteria disclosed before dispatch; these are not official upstream issue grades. Known edit allowlists disclose six/seven files. Project implementation sizes are 10,592/1,614 lines; this is not an unhinted large-repository efficiency test.

Only `light-edit.js` differs between frozen Light arms; actual initial contracts and normalized prompts match. Whole-symbol replacement accepts one exact existing JS/TS/JSX/TSX first-line whitespace prefix without changing later bytes, signatures, exports, inline declarations or edit guards. Six byte/safety cases improve from three to six passes; replaying the original F30 symbol batches changes repeated indentation from $2 \to 4 \to 6$ to $2 \to 2 \to 2$. This is a known-input mechanism check. The preceding 283 focused regressions pass.

Arm	Official	Frozen endpoint	Exact checks	Full audit
Reference Light	3/6	6/6	3/6	3/6
Modified Light	4/6	5/6	3/6	3/6
Integrated Cline	4/6	6/6	5/6	2/6

Endpoint acceptance includes new probes, types and original suites, including fast-glob's separate E2E suite. The modified arm's adverse endpoint passes feature tests but fails an existing stream-reader mock test. Five Light trials reach 32 iterations; two native trials stop at the research request-byte guard before dispatch. Correct independently graded sources do not recover missing delivery.

The table uses the superseding F33 audit. The earlier F31 reviewer correction mistook returned Rollup/tsc script labels for the executed process. Frozen code already launches `npm run SCRIPT`, preserving lifecycle semantics. F33 restores four full-delivery judgments. The literal `npm test` requirement still differs from the executed `npm run test`: one previously accepted check-only judgment is rejected, and another failure retains this narrower reason. Both preserve hooks; do not infer bypassed lifecycle execution. A note correction retains the wrong-order failure. Original F31 judgments, correction chains, invocation bindings, sources, scores and resource bytes remain unchanged. The old resolved-project sensitivity rests on that mistaken premise; this is now a correction under the same exact-wrapper criterion, not a relaxation. A separate reporting-only sensitivity admits one native count-report error; a final-diff-only sensitivity admits one reverted source-scope violation. These relaxations are not combined. Another stop-cause note correction changes no grade. Actual preregistration uses the launch-bound commit/time; the fixed `taskInputCommitAtUTC` is a study-date label, not a measured commit timestamp.

Other current friction is concrete: dependency navigation refuses installed internals, then agents use shell reads; a scoped multi-file patch remains refused until an explicit route correction; native dependent commands can be submitted concurrently despite an explicit independence contract. One native YAML check fails before build completion and its final layout diagnosis is

unsupported. Another native delivery assigns E2E totals to the unit command. See the SHA-bound call/source evidence and current-code status in the phase's `findings.md`.

All 12 Light trajectories have zero whole-symbol replacement exposure. The registered quality/resource gates still fail. Three strict modified/reference YAML pairs have −5.0% tokens, +1.9% estimated cost and −10.7% worker time. Neither these contrasts nor all-attempt totals identify an effect of inactive indentation handling. Retain the bounded byte-correctness repair after tracing the adverse outcome to the model's stream patch; preserve the failed gate and the outcome. This is not population noninferiority or a model optimization endorsement. Public reconstruction of all18 original endpoints matches source hashes, builds, UTC/supplied probes, types, original default and E2E suites, including the known failure. Non-UTC original scores remain separate; no source recovery or model rerun occurs.

10 F32: isolated validation presentation, not promoted

The prototype indexes detected raw Node/Jest/Mocha/QUnit summary lines hidden in the middle of long validation logs. Full-log line locations, explicit partial markers and "not verification" labeling avoid aggregating stages or inferring zeros. The existing 900-character normal and 4,000-character failure/fallback limits remain; short output up to 1,200 characters and appended Node-event captions retain their prior behavior. Logging safeguards, execution, counts, source/environment binding and admission logic are unchanged. Eight controls improve from two to eight passes, including real successful/failed commands, unknown pending, unverified admission and symlink-safe fallback. The temporary integration passes 581 current-source harness cases with two skipped (583 total), compilation and lint. Two failures from an abandoned compiled F29 prototype reproduce on the old runtime; they are not current-source regressions. Setup loader/pnpm failures are retained separately.

However, all52 detected summary occurrences are already visible in all17 selected F31 Light logs, including failed trials: no output changes, no gained/lost lines and no missing logs. The fixed-budget outputs remain 42,531 characters /43,707 UTF-8 bytes. A local 50-repetition alternating-order `process.cpuUsage` measurement adds about 0.139ms per log for scanning. This is not model latency or billed tokens. Revert the temporary production integration and keep the candidate, controls, public logs/replay and rollback evidence isolated. Constructed visibility gains do not override this neutral observed replay. No F32 model calls or model efficacy measurement are claimed; frozen F31 runtimes remain unchanged. Restored production compiles, passes all88 validation/indentation controls and lint, and matches the original presentation module hashes.

11 F33: truthful invocation display and audit correction

Project tasks now render the actual executable and argv, including managed reporter arguments, rather than the underlying package script. POSIX-style quoting preserves literal argument boundaries as diagnostic notation; execution still uses structured argv. Raw caller commands remain exact. Catalog definitions still identify the runner. No process, lifecycle or verification-admission change is intended.

On identical real-process controls, display-to-spawn agreement improves from 0/3 to 3/3 and new regressions from 1/3 to 3/3. All six npm hook events remain identical; counts and admission do not change. Public reconstruction of old modules reproduces the result. Compilation and lint pass; 578 current-source harness cases yield 576 passes and two skips. Seven corruption controls reject fabricated acceptance/resources, removed hooks, changed executable and promoted admission. An initial research-verifier path-normalization failure is retained as setup, separate from runtime results. Historical invocations are reconstructed from frozen code and recorded inputs, not historical spawn logs. The superseding audit above does not measure a runtime improvement. Retain the display repair for factual correctness; model efficacy remains a separate

experiment. Replaying all32 successful recorded project labels reduces the command field from 797 to 495 UTF-8 bytes (21 shorter, 11 longer). This is neither complete tool output nor billed-token savings, model latency or a comparable-quality efficiency gain.

12 F34: isolated display diagnostic, no model gain

Nine precommitted assignments reuse the known YAML task, three dependent repeats and three rotated arms. Only the three command-rendering modules differ; all six normalized initial Light contracts match. This is not fresh confirmation. The inherited task's "fresh-confirmation" tag is stale; the preregistered phase classification already states known diagnostic. Original protocol bytes remain.

Arm	Official	Frozen endpoint	Strict full audit
Reference Light	3/3	3/3	3/3
Display-fix Light	2/3	3/3	2/3
Integrated Cline	2/3	2/3	1/3

An exec-server transport interruption kills the first candidate worker after three successful checks and a started-only fourth. Preserve its missing result/time, partial usage and reservation; capture dead-worker state and unchanged source, then resume only the seven unexecuted slots. No model trial is retried. Cline's first reported response uses 128,000 output tokens and malformed `parallel` arguments, then hits the USD0.10 soft-cost stop after546.474s; this is not the600s timeout. Its second recorded forwarding attempt lacks usage. Another native trajectory has correct source and visible passing tests but a semicolon chain leaves earlier individual non-test exits unrecorded. Strict review withholds acceptance; an explicit apparent-validation sensitivity gives3/3,2/3,2/3. F34 did not separately bind F31's exact-or-`&&` rule: this is reviewer uncertainty, not a user ban on semicolons. Note corrections remain public.

Two strict candidate/reference pairs show +19.8% tokens, +26.8% estimated cost and +16.0% worker time. Both gates fail. Candidate R2 has no changed-label exposure; R3 has two. A single strict native pair shows −30.3% tokens and −24.6% cost but +12.0% time for candidate. These small dependent contrasts identify neither population superiority nor a causal display effect. Retain F33's locally proved factual repair, without a model-optimization claim.

All91 recorded forwarding attempts omit the nominal8,192-token output limit. Application configuration and actual Core config splitting preserve8,192; 17 existing configuration/legacy-handler controls pass. Those controls do not exercise the complete SDK start/model-selection/request path; their test description now states that narrower boundary. At F34 completion the drop layer, backend support and reasoning portion remained unknown. F35 addresses that separate question.

13 F35: effective output-limit contract

Full public SDK offline capture reproduces the omission at the Codex host while the same configuration reaches an ordinary endpoint. The installed provider module explicitly suppresses output limits for `chatgpt.com`. Before production changes, commit `b3618c2a` registers one live output-stress request capped at 1,024. HTTP 200 and `response.incomplete`/`max_output_tokens` report exactly 1,024 output tokens, including 398 reasoning tokens. The SDK fails the incomplete turn as expected; this is enforcement evidence, not user-task success.

Commit `fb9b1b36` adds the opt-in HTTP correction described above. Cap-specific controls improve 3/8 to 8/8, and full SDK configured/unconfigured/other-origin controls 2/3 to 3/3. Compilation, changed-source lint and 108 targeted cases pass. The registered runtime suite yields 875 passes, two skips, zero failures across 877 cases. Corrected offline-fixture/setup errors remain

disclosed. Retain contract correctness on the tested endpoint/model; useful-output truncation remains a quality risk, and backend support is not established for every provider/revision.

14 F36: actual-cap diagnostic, completed

Commit `5cf2656b` preregisters 12 assignments on the known YAML task, three dependent repeats and four rotated arms: uncapped/capped Light and uncapped/capped integrated Cline. Only the HTTP helper differs between frozen runtimes; factory wiring is common. This is neither a literal old release checkout nor fresh-task confirmation. All six initial normalized contracts agree within each harness.

Arm	Official	Frozen endpoint	Strict full audit
Reference Light	3/3	3/3	2/3
Capped Light	3/3	3/3	2/3
Uncapped integrated Cline	1/3	2/3	1/3
Capped integrated Cline	3/3	3/3	2/3

Both observed quality gates pass. Each strict paired-resource comparison has only repetition one: capped/reference Light has +45.1% tokens, +1.0% cost and +22.0% worker time; capped/uncapped native has −24.2%, +4.6% and +10.5%. Both resource gates fail. Capped Light/native has −20.9% tokens, −33.1% cost and −7.8% time on one joint success; this is not general superiority. Across all assignments capped Light is worse on all three resources. Capped native worker time is 77.1% lower, but two missing uncapped usages preclude complete token/cost comparison. Keep these denominators separate.

All 72 treatment forwarding attempts carry 8,192 and all 51 references omit it. There are 121 usage reports for 123 attempts. No reported treatment output reaches the cap; the maximum is 2,044. This does not exclude a parameter-driven planning effect, but no 8,192 truncation mediator is observed. Uncapped native R2/R3 time out at 600.875s/601.086s: one has a passing endpoint without agent verification or delivery; the other leaves the feature absent. SDK zero accumulators and its `context.modelRequests` report counter do not prove zero attempts/spending. Original values remain; the wire/report ledger and reserves preserve uncertainty.

Other full-audit failures retain a reverted forbidden source edit, literal npm invocation/reporting mismatch, and capped native R3's earlier semicolon-chain exit uncertainty. The latter is reviewer uncertainty, not a user ban; the separate apparent-validation sensitivity gives capped native 3/3 without replacing primary grades. Both npm spellings retain lifecycle hooks. Current scope advice already exists despite repeated refusals; it must not be counted as an absent contract. Retain explicit-budget correctness, not an optimal-cap or population-efficiency claim. The phase README, original archives and `findings.md` contain the call/source evidence and current-code status.

15 Accounting, decisions and reproduction

F28/F29 used cumulative EUR 10 authorization, without resets. For this finite F30/F31 phases the runner voluntarily retains that technical limit; the user requests continuation without further budget questions. Conservative reference USD/EUR 1.11751153 and 2% headroom give USD 10.951612 operational stop. F28/F29 use per-trial soft stop USD 0.04, USD 0.23 final-response reserve, 24 iterations and 480s timeout. F30/F31 fix USD 0.10, the same reserve, 32 iterations and 600s, with a phase USD 6 stop. All retain 290,000 serialized request bytes and a 25s outer watchdog. The nominal 8,192 output cap was absent in F28–F34 requests; do not retroactively treat it as effective.

F29 adds observed USD 0.24125812 and no new missing-response reserve. The F29 cumulative observed estimate is USD 2.82173048 plus USD 0.92 retained reserves. F30 adds USD 0.54532960, with no new consumption gaps or reserves. The new cumulative observed estimate is USD 3.36706008 plus USD 0.92 reserves: USD 4.28706008, reference EUR 3.83626. Reserves are not imputed spending; costs are SDK estimates, not invoices. Cache reads are a subset of input, reasoning a subset of output; full per-request cache, hidden retries and researcher-agent/host/setup/ tool/oracle costs remain unestablished. All failures, missing consumption and timing uncertainty remain.

F31 adds estimated USD 0.74957032 with no new wire/usage-count gap or reserve. Cumulative observed estimates become USD 4.11663040 plus the retained USD 0.92 reserves: USD 5.03663040, reference EUR 4.5070. Aggregate cache reads are recorded but full per-request cache and reasoning accounting remain incomplete. F32 local replays add no paid model requests; host/researcher/tool costs remain unpriced.

F34 adds known estimated USD0.313557, including USD0.01252348 in nine usage events from the interrupted trial, plus USD0.56 retained reserves (missing aggregate0.33, unreported native attempt0.23). The full missing-aggregate allowance remains in addition to observed partial usage; conservative overlap is explicit. The original scheduler ledger is retained separately. Cumulative observed estimates are USD4.4301874 plus USD1.48 reserves, accounted USD5.9101874. Missing aggregate/time, the wire/report gap, incomplete cache/reasoning and unpriced work remain explicit. F35 adds estimated USD 0.0018922 with no new reserve: cumulative observed USD 4.4320796 plus USD 1.48 reserves, accounted USD 5.9120796. Backend reasoning is observed in that probe but remains unknown in its SDK-normalized usage. F36 adds estimated USD 0.25372472 and USD 0.46 reserves for two unreported requests, accounted USD 0.71372472. Cumulative observed estimates become USD 4.68580432 plus USD 1.94 reserves, accounted USD 6.62580432. A zero SDK accumulator without a usage receipt is not measured zero consumption. Cache/reasoning and invoice completeness remain unestablished; all original timeouts and receipts are retained.

Retain conservative correctness/uncertainty contracts and the locally validated diagnostic repairs. Do not sell the six-change bundle as an optimization or activate the ranking candidate from lower token totals. F30 also rejects the concise-wording prototype. These studies improve the reliability of the evidence by exposing false acceptance; they do not prove new global harness superiority.

Artifacts and public no-model replay commands are in:

- `docs/experiments/light-harness-guidance-f28-2026-10-15`
- `docs/experiments/light-harness-phrase-ranking-f29-2026-10-15`
- `docs/experiments/light-harness-concise-f30-2026-10-15`
- `docs/experiments/light-harness-edit-boundaries-f31-2026-10-16`
- `docs/experiments/light-harness-validation-view-f32-2026-10-16`
- `docs/experiments/light-harness-invocation-f33-2026-10-16`
- `docs/experiments/light-harness-invocation-model-f34-2026-10-16`
- `tools/terminal_bench/guidance-study/verify-public.py`
- `tools/terminal_bench/phrase-study/verify-public.py`
- `tools/terminal_bench/request-limit-study/verify-public.mjs`
- `tools/terminal_bench/request-limit-model-study/verify-public.mjs`

Public byte/accounting replays verify original compressed bytes/hashes, recorded manual decisions, counterexample arithmetic, paired resources, wire settings and cumulative reserves. They do not independently re-adjudicate source or execute JavaScript oracles. Negative controls reject changed outcomes, costs, pairs, reserves, model settings and supplemental behavior decisions. Separate Node scripts reproduce local probes; F30 also reconstructs and probes public source endpoints without private workspaces or model access. Its reporting-only sensitivity never replaces the primary gate. F36's full public-source replay reconstructs all 12 endpoints, preserves recorded source hashes and matches build, UTC/non-UTC probes, supplied tests, types and original suite outcomes, including the unchanged-source failure. This reproduces outcomes, not agent actions or twelve successful tasks. All 12 publication corruption controls pass, including rehashed request-setting changes; F35 has seven separate controls.

16 F37: read-only dependency declarations, local mechanism

Following the F36 audit, the iterative cycle resumed with one navigation change. Its protocol was committed as `9977de17` before implementation; production reference `8ceb9139` is retained as a SHA-bound compiled archive. F31's known fast-glob traces (zero-based baseline tool indices 22–24, candidate 24–25) refuse dependency declarations, then use shell discovery. This exclusion persisted in `readPath`; configured compiler references/rename were already implemented.

Implemented. `read\_code` now accepts explicit `view: declaration`, anchored in `localFile\#importBinding.member`. The existing bounded compiler worker resolves aliases/re-exports and returns installed package/version identity, source/package hashes, project scope/warnings, overloads or bounded namespace exports. Content pages contain at most 12,000 UTF-8 bytes, with byte-offset continuation; at most 16 declarations/256,000 projected bytes are admitted. Canonical package ownership, protected subpaths and realpath/content freshness cover pnpm/shared installations and symlink replacement. Ordinary dependency reads/edits remain refused. This grants no writable alias, edit coverage, rename authority or validation admission. Missing configuration/types, ambiguous imports and local augmentations fail explicitly. Type declarations do not establish dependency runtime behavior.

New local measurements. Six known fast-glob lookups, repeated three times with rotating arm order and fresh sources/sessions, recover correct identity in 18/18 candidate reads versus 0/18 reference refusals. Independent installed-file/package hashes agree. The configured project contains 79 roots/221 loaded files. Response bytes rise from 4,023 to 22,434; total lookup time rises from 1.049s to 16.867s. Short refusals deliver less information, so these are not resources at comparable quality. They are dependent diagnostic lookups, not eighteen tasks, billed-token savings or model-success evidence. SDK JSON tool definitions add 516 bytes entirely in `read\_code`, including when unused; other workspace definitions are byte-identical. This is not billed tokens or a wire/cache measurement. No model requests or additional cost occurred. The first test panel passes 12/15; three fixture mistakes are corrected (ESM identity, JS/TS source shadowing and an existing refusal status). Expanded targeted suites pass 52/52 including sixteen new controls. Build/lint and registered-suite outcomes, original logs and replay scripts are published in `docs/experiments/light-harness-dependency-declarations-f37-2026-10-16`.

Decision at the local phase. Retain the explicit read-only capability with verified identity/boundaries; global utility remains unproven and compiler startup/context overhead is adverse. New unhinted cross-file tasks on a larger project must compare reference Light, declaration Light and integrated Cline under identical actual model/effort/caps and frozen budgets. This local phase has not registered or executed that model panel. F38b below subsequently executes that comparison without startup caching.

17 F38–F41: four completed iterative cycles

Protocols and ordered assignments precede outcomes; each model treatment remains isolated. Original traces, adverse controls, scripts and compact phase reports are in `docs/experiments/light-harness-iterative-f38-2026-10-16` and `docs/experiments/light-harness-compiler-check-model-f41-2026-10-16`.

F38b: new-project model comparison, adverse result. Axios v1.12.2, commit `4150efd104826d7af27d014378bf874964aa26ff`, has 62 production JavaScript files. Two researcher-authored tasks require interceptor clear counts or strict asynchronous legacy FormData length, with implementation and ESM/CJS public types. Prompts omit implementation locations. Selected supplied runtime/regression/type checks have before/positive controls; the full upstream browser/package suite is not run. Common research JS configuration enables installed-dependency declarations; it is not the upstream default. Two dependent repeats rotate pre-F37 Light, declaration Light and integrated SDK. All actual output caps are 8192, with 32 iterations, 600s plus 25s watchdog, 290,000-byte requests and SDK USD 0.10 soft limit plus USD 0.23 reserve. Source/dependency locks, executable trees, prompts, schemas, wire observations, checks and budgets are frozen.

Phase	Arm	Audited	Tokens	SDK USD	Seconds
F38b	Reference	4/4	558,003	0.045514	231.656
	Declarations	4/4	701,178	0.058537	250.641
	Integrated Cline	4/4	814,676	0.069509	419.105
F41	Reference	3/3	440,959	0.036646	145.353
	Compiler admission	3/3	349,964	0.027621	106.480
	Integrated Cline	3/3	489,383	0.042915	128.526

Official runtime and independent selected-check successes equal audited counts in both panels. Review is unblinded by the research agent, without independent human adjudication or solution repair. No declaration-view call occurs in any F38b trial. Candidate/reference tokens rise 25.7%, estimated cost 28.6% and time 8.2%; its quality gate passes and resource gate fails. No declaration mediation or general utility gain is established. Retain the bounded capability on local correctness evidence; reject its demonstrated-global-benefit claim on this panel.

The separate initial F38 launch is infrastructure-invalid: a JS-only archive lacks prompt assets, three Light initializations fail, one native task runs and eight slots never start. Original four traces, missing fourth scheduler exit and USD 0.01284832 observed plus USD 0.99 conservative reserves remain separate. F38b restores identical common assets and freezes again. The native clear task was already observed, so F38b is not wholly fresh confirmation. Oracle/configuration setup failures are retained rather than attributed to the model or Light quality.

F39: implemented scope diagnosis, local evidence. F36's known refused patches expose exact whole-file context beyond the selected symbol. After scoped context mismatch, a bounded nonmutating planner now returns affected-line hints only when the full-file patch is unambiguously applicable. It never applies/widens the patch or grants observation coverage. Limits are 512 KiB source and 32 KiB patch; stale, ambiguous and malformed cases retain guards. New controls improve 6/10 to 10/10; relevant suites pass 32/32. Seven original refusals replayed against the exact observed preimage remain blocked: three gain the specific scope diagnosis, four retain existing header diagnoses. Returned bytes rise 10,015 to 10,855; noisy local time rises 1.748s to 2.545s. Keep the correct diagnosis; fewer model retries or tokens are not demonstrated.

F40: implemented honest request observations. Historical `modelRequests` counted input-usage receipts: native F36 R3 has one wire attempt but zero reports; R2 has four attempts and

three reports. The event-only collector now returns unknown transport attempts/completed responses and nullable input means/maxima, with separately labeled iteration-start, usage and input-token report counts. An actual zero remains zero. New controls improve 0/8 to 8/8; relevant suites pass 15/15 and the registered runner passes 37/37. Four original trace replays preserve other metrics and accounting. Legacy report consumers retain historical behavior and explicitly reject the new unsupported semantics (four controls). Keep truthful unknowns; transport instrumentation and provider billing are not supplied by this repair. No task-success gain is claimed.

F41: observation-driven compiler admission. F38b's eight Light trials run the required compiler with exit0, yet script-name classification treats `test:study:types` as a test and leaves its obligation missing. The narrow fix recognizes literal compiler-only `tsc -noEmit` forms and preserves npm lifecycle/actual process status. New records/journals explicitly carry `invocation_only` scope and no inferred source paths. The first classification-only repair passes eight controls but falsely verifies an edited source excluded from the compiler project. That candidate is rejected; the scoped repair passes nine new controls and 55 relevant cases. This does not repair older project-scope heuristics for all other build/type tasks. F44 below subsequently removes name-based scope admission for every catalogued typecheck; other build/lint heuristics remain a separate boundary.

Nine model trials on the known Axios clear task rotate three dependent repeats. Reference already includes F37; only four compiler/journal modules change. F39/F40 are excluded from these frozen snapshots. Required compiler admission improves 0/3 to 3/3, while all six Light source-verification results remain `unverified`; native structured admission is N/A. One additional native Mocha command fails127 because upstream `cross-env` is absent, is disclosed, and is retained. Required selected checks pass; no unrun upstream suite is claimed.

Both F41 registered aggregate gates pass: compiler/reference tokens fall 20.6%, estimated cost 24.6% and duration 26.7%; versus Cline they fall 28.5%, 35.6% and 17.2%. Token efficiency is fragile: paired changes versus reference are +15.5%, −50.3%, +22.7%. Omitting retry-heavy reference repeat2 reverses the aggregate to +19.0%; this is a labeled post-hoc sensitivity, not a replacement cohort/gate. Keep the correctness fix, deepen efficiency confirmation. Cache and stochastic editing trajectories prevent causal resource attribution here.

Reproduction, validation and accounting. All 21 public endpoints reconstruct from pristine source plus original agent diffs and reproduce 63 selected script outcomes, without model calls or solution recovery. These are endpoint reproductions, not 21 additional agent tasks or admission replays. Aggregate input/output/cost receipts reconcile, but per-request cache details remain incomplete. SDK estimates are not invoices; cache reads are included in input tokens and unknown reasoning details are not added again. An existing 50ms cursor test flakes in the first runtime suite; a separate test-only commit synchronizes output chunks without changing process implementation. Final registered runtime: 910 passed, two skipped, zero failed; runner: 37/37. Before-fix and rejected-scope failures remain published. Total new observed SDK estimate is USD 0.29359032 (initial F38, F38b, F41); cumulative historical/new observed is USD 4.97939464 plus unchanged USD 2.93 reserves. Campaigns are never pooled.

18 F42: fresh-task confirmation, completed

The preregistered protocol, all assignments, raw traces, source reviews and reproduction scripts are in `docs/experiments/light-harness-confirmation-f42-2026-10-17`. Six constructed multi-API requests span previously unmeasured Valibot v1.1.0 (508 production TS modules, 38,848 lines) and Remeda v2.30.0 (208 modules, 16,150 lines), with three dependent repeats and three arms. Two configuration requests share machinery; these are not 54 independent issues. Current Light differs from its reference only in literal compiler recognition; F39/F40/refactors are common.

Integrated SDK includes the common native-search repair. Frozen model/effort, 8192 output tokens, 48 iterations, 900s plus a 25s watchdog, 500,000 request bytes and USD0.25 soft session plus USD0.23 reserve remain fixed; panel/cumulative scheduling ceilings are USD30/100.

Outcome /18 assignments	Reference	Current	Cline
Official completion	18	18	17
Selected independent checks	18	18	17
Strict audited completion	10	9	7
Source and delivered-test audit	11	13	13
Optional config-shape sensitivity (source only)	14	16	15
Append-only test-placement sensitivity	11	12	10
Required compiler admission	0	18	—

The strict primary reading preserves existing tests. Adding new cases to an existing test file is plausibly permitted by "own new tests"; report that sensitivity without calling additions weakened assertions. Eight config endpoints add an own undefined option, failing original exact-object tests; this also affects F42 positive controls and is interpretation-sensitive API evolution. The separate shape sensitivity accepts it without forgiving other errors. Uniform post-hoc witnesses run unchanged sources for all relevant endpoints: three numerical regressions, one broken existing two-type-argument API, three invalid added-test sets, and an extra website type defect remain. They are not new model successes, repaired solutions or changed official checks. Truthful delivery is separate: one Cline formatter claim and one Light claim about the chronology of an extra check are unsupported. Three required checks can pass while extra delivered code/tests or broader assertions fail.

On six paired strict successes, current/reference tokens are +10.1%, SDK cost +6.9% and observed duration -21.4%; token leave-pair-out results include both signs. Against Cline only two strict pairs remain, with +46.2% tokens. Both registered gates fail, including the minimum twelve paired-success requirement. Retain compiler admission for its demonstrated recognition correctness, not as proof of improved task success or efficiency. No arm admits whole-source verification. Declaration reads remain unused; three reference searches use TypeScript identity, thirteen fall back for a missing symbol target. Light makes 31 missing-path reads, twenty with original-tree basename candidates; that inventory is an opportunity estimate, not an intermediate-state replay.

Cline assignment36 vanished with its driver before an edit/final result; cause and exact time are unknown. It remains a failure in assigned denominators, with no retry and null full consumption/duration. Four partial receipts report USD0.01216492 separately. Completed aggregate SDK estimates total USD1.58475536; new reserves are USD0.71. Cumulative observed/reserved estimates are USD6.56415/3.64, not invoices. Cache reads are included in input; request-level cache reconciliation is incomplete. Warm auxiliary caches, reachable ancestor tools and recorded validation-load overlap limit timing/environment conclusions. Generated cache drift stopped the driver before assignment6; original inputs were restored and only untouched slots resumed. No result was dropped.

19 F43: isolated path-recovery follow-up, completed

The bounded missing-path candidate returns paths only, preserves ambiguity and requires an explicit read. Ten new controls cover stale indexes, cancellation, alias collisions, protected/symlink files and edit coverage. Nine distinct known failed inputs from F42's first ten assignments yield five valid suggestions versus zero on reference, on pristine inputs; this is not a conversation replay or a token saving. Full runtime validation passes 920 tests with two skipped; runner 37/37. A pre-existing Markdown formatter fixture failure reproduces on both reference and candidate with

pinned pnpm; a separate test-only repair makes the mocked formatter actually edit Markdown. Original failures are retained.

The registered 18 assignments use three known Valibot requests, two dependent repeats and three arms. Only `light-workspace.js` differs between Light executables; compiler admission, helpers, prompts and schemas are common. All slots complete without retry or timeout. Per-trial copied dependencies, empty offline caches, pinned npm11.6.4/login-shell PATH, related original tests, old explicit-generic fixtures and clearer existing-test preservation are common to every arm. These differ from F42; cross-campaign quality changes cannot be attributed to recovery. No other model/test/compiler load is scheduled during workers; the host is not exclusive and no OS sandbox is claimed.

Arm	Audited	All tokens	SDK USD	Seconds
Reference	6/6	2,359,099	0.156990	633.384
Recovery	6/6	2,899,913	0.166531	692.165
Integrated Cline	5/6	4,115,964	0.238883	912.807

Official and selected-check successes are 6/6 for all arms. Uniform original runtime/type audits pass on three pristine controls and all 18 endpoints; one Cline added-test helper fails TypeScript with upstream exact optional properties, although its five runtime tests pass. Its production implementation passes. Uniform extra website metadata checks pass for all endpoints. All 18 original-patch selected-check replays match, as do the public audit replay of three pristine controls and the failing added-test endpoint. Six corrupted-ledger controls are rejected. No solution is repaired. Neither Light arm admits whole-source verification; both admit six required compiler invocations. Review remains unblinded and focused, not exhaustive.

Three feedback suggestions are followed in later iterations, two as the direct next navigation; total feedback is 1,602 serialized bytes, not provider tokens. These occur only in the two message-prefix trials, whose paired tokens fall 27.4%. Hide-input rises 56.5% and fallback-when 42.5%, with no recovery offer in those tasks. Across six audited Light pairs, tokens rise 22.9%, SDK cost 6.1% and duration 9.3%; three pairs win and three lose. Leave-pair-out token results span -7.5% to +42.2%; every leave-task-out result remains adverse. Quality and observed-use gates pass, but the registered efficiency gate fails. Retain the bounded diagnostic on local correctness and observed use; reject its demonstrated-efficiency claim. Unused feedback cannot explain costly trajectories.

Against Cline on five paired audited successes, candidate tokens/cost/duration fall 42.6/38.3/34.5%. Keeping all six selected-check successes gives a smaller 29.5% token difference, but includes the invalid delivered test. All costs and failures remain in the table; this panel does not establish general superiority. Declaration reads remain unused and all six Light reference searches fall back lexically for a missing symbol target. Native `run_commands` carries unsupported timeout fields in 122/132 calls; the captured schema exposes commands only and SDK source uses a factory timeout. This is a source/schema mismatch, not an observed timeout failure or a timeout repair in this cycle.

All 274 recorded forwarded attempts have usage reports; input/output/SDK-cost reconcile, but full per-request cache does not. Cache reads are included in input; zero aggregate cache-write is not proof of zero writes. F43 adds estimated USD0.56240384 and no reserve; cumulative completed-aggregate observations are USD7.12655384 plus retained USD3.64 reserves. F42 partial receipts remain separately disclosed. Invoices, hidden retries and researcher/host/tool/audit costs remain unestablished. Original patches, traces, uniform witnesses, independent integrity checks and no-model replay commands are in `docs/experiments/light-harness-path-recovery-f43-2026-10-17`.

20 F44: compiler membership feedback, completed and withdrawn

A current local defect is independent of the historical results: the same successful TypeScript command excluding an invalid changed test falsely admits source verification when called `typecheck`, but not when called `test:study:types`. The retained catalog guard marks all typecheck tasks `invocation_only`. Both spellings now satisfy execution obligations without verifying the excluded file. The guard is common to both Light arms; its local correctness gain is not the effect measured below.

The isolated prototype appends `-listFiles -pretty false` to supported literal local npm no-emit compiler invocations, never `-listFilesOnly`. It reports bounded changed-file membership and compiler/source/manifest identity, not runtime coverage, type correctness of every listed file, installed-environment identity or admitted verification. Unsupported hooks/wrappers, incomplete output, identity changes and stale source remain unknown. No check is silently widened. The frozen executables differ at one capture call site; prompts and schemas match.

Three new constructed requests on previously studied Valibot and Remeda require new regression files: parser per-call config, summary formatting, and eager/lazy take-count normalization. Three dependent repeats and three arms give 27 fixed, rotated assignments, with no retries. Common pinned dependencies, isolated offline caches and four selected checks are supplemented by independent unchanged-patch runtime/type checks of all added tests. Model/effort remains `gpt-5.6-luna/high`; per trial USD0.25 SDK soft limit plus USD0.23 reserve, 900 seconds, 48 iterations, 8,192 output tokens and 500,000 request bytes. The panel allowance is USD12. Initial positive-fixture errors and two stale runtime test expectations are retained separately from product regressions.

Arm	Official	Audited	All tokens	SDK USD	Seconds
Reference	9/9	6/9	3,380,764	0.222346	1,288.107
Membership	8/9	6/9	5,240,047	0.296118	1,581.686
Integrated Cline	9/9	7/9	4,932,928	0.303927	1,395.195

All 27 endpoints pass the four selected checks. One reference added test runs but fails strict compilation (four TS2339 errors); the other 26 added-test sets pass both audits. Uniform original take/pipe tests pass on pristine input and all nine Remeda endpoints. A post-hoc scalar-result witness nevertheless fails on six unchanged patches: a new empty-sequence branch bypasses the single-result conversion, returning an array instead of undefined after `first()`. The original input and three patches pass. This is a disclosed diagnostic, not a fresh confirmation task, repaired solution or changed official score.

The prototype exposes 25 missing-file reports in all nine trials; successful broader checks follow in later iterations in eight trials (nine commands). The frozen counter only recognizes named delivered scripts and reads zero; hash-bound manual chronology additionally captures direct Vitest/tsc and a full package run. These are temporal observations, not causal mediation. One follow-up finds and repairs added-test type errors. Conversely, one correct prototype patch claims type coverage from runtime-only Vitest, failing strict truthful delivery; source-only sensitivity is 7/9 versus reference 6/9. A separate real-runner control confirms that Vitest's "Type Errors: no errors" can coexist with TS2322 in an excluded runtime test. Reporting and acceptance cannot be inferred from the green summary. No Light arm admits whole-source verification.

On six paired strict successes, prototype/reference tokens increase 12.5%, SDK cost 17.8%, and observed duration falls 14.5%; two token wins and four losses, with leave-pair-out effects from -4.5% to +48.0%. Across all assignments, tokens/cost/duration increase 55.0/33.2/22.8%. Against Cline, five strict pairs show -5.9% tokens, -6.6% cost and +13.0% duration; all assignments instead show +6.2% tokens and lower audited quality. Neither comparison establishes general efficiency. The registered default-retention gate fails: withdraw membership feedback, preserve its frozen

prototype/results, and retain the independent typecheck scope guard. Forgiving the inaccurate delivery would change a small quality count; it does not replace the registered strict decision.

One prototype trajectory reaches 48 iterations after four rejected finishes: an optional root-relative test selector, executed from a child-package cwd, finds no files. Later full-package passes cannot clear the historical selection's code-error classification. Its 2,017,701 tokens remain in all totals; this shared pre-existing mechanism was exposed by the treatment, not excused from its score. The next phase treats it separately. All 452 forwarded attempts have receipts; input/output/SDK cost reconcile, request-level cache does not. Aggregate cache reads are included in input and zero cache-write does not establish zero writes. F44 adds estimated USD0.82239108 with no new reserve; cumulative completed observations are USD7.94894492 plus USD3.64 retained reserves. Hidden retries, invoices, researcher/tool/audit costs and external host contention remain unknown. All 27 archived patches reproduce their six original endpoint-check outcomes, including the failing added-test compilation, without source repair. Evidence and replay commands are in `docs/experiments/light-harness-validation-scope-f44-2026-10-17`.

21 F45: test-selection recovery, local evidence

After all F44 workers finish, a separate limited correction recognizes a single supported Vitest no-files diagnostic with exit one and no conflicting test, compiler or setup failure. It classifies an unavailable command selection, displays its effective cwd, and preserves the failed invocation. Counts stay unknown, source verification stays unadmitted, and a required exact selection remains missing. An unchanged unavailable selection is not automatically retried; stale source is reconsidered. Actual failed tests still block after a different filter passes. Unknown versions, cancellation and contradictory output keep conservative handling. Text diagnostics remain observations, not coverage proof.

The unchanged failing F44 Remeda patch is reconstructed with the same bad selector, four required checks and the full package test task. Frozen reference rejects both repeated finishes; corrected runtime permits an unverified finish with zero automatic retries, retaining the failed selector. Making that exact selector required leaves its obligation missing. More than 3,800 tests pass in both arms, yet the separate scalar witness still fails: no source repair or additional audited/model success is claimed. Nineteen new controls cover the selection correction; full runtime validation passes 941 tests with two skipped, and runner validation passes 37/37. Changed-runtime lint and repository line budgets pass. Independent replay receipts are archived with the phase; F47 below separately measures its optional use on new requests. Evidence and no-model commands are in `docs/experiments/light-harness-selection-recovery-f45-2026-10-17`.

22 F46: formal admission checks, local evidence

An offline model compares 1,536 prespecified journal/admission states against the real current journal and finish decision: all match. Nine Z3 witnesses distinguish actual scratch-code mutations; 18 latest-invocation and six non-mutating controls also match. The adapter bypasses automatic scheduling to preserve fixed obligations, so this is admission-state verification, not an end-to-end runner or application proof. No existing defect is found within the abstraction. Scalar case generation takes 0.0032 seconds versus 0.0275 for nine solver checks in one local observation, with different setup boundaries: no solver speedup or task-quality gain is established. Keep this optional development study; simple runtime predicates remain in TypeScript. Evidence: `docs/experiments/light-harness-formal-validation-f46-2026-10-18`.

23 F47: selection recovery, new constructed requests

Three new multi-package workflow requests (correlation IDs, CSV options and token expiry) receive two rotated repetitions in each arm. Frozen reference predates F45; candidate differs in its four production modules only. F48/F49 are absent. No bad selector, recovered solution or solver query is supplied. All 18 official, selected, independent functional and strict reviewed outcomes pass; these remain separate predicates even where their counts coincide.

Arm	Strict	Tokens	SDK USD	Seconds
Reference	6/6	496,873	.068124	522.3
F45	6/6	537,920	.069154	366.4
Integrated Cline	6/6	727,120	.117718	1,122.0

No empty selection occurs: mechanism use is zero. Candidate/reference tokens rise 8.3%, cost 1.5%, while observed duration falls 29.8%; the registered resource gate fails. Leave-pair-out token effects range from -1.8% to +15.1%, and leave-task-out from -9.4% to +19.3%. Candidate/Cline uses 26.0% fewer tokens and 41.3% lower SDK cost, but correlation alone is 32.6% worse in tokens. Neither comparison attributes gains to unused recovery. Retain F45's local correctness fix without claiming a demonstrated model benefit.

All 161 forwarded attempts have usage receipts; cache attribution per request does not reconcile. Tokens include cache reads, and reported zero writes do not prove none occurred. The campaign adds estimated USD0.25499584; cumulative observed SDK estimates are USD8.20394076 plus USD3.64 retained reserves. Long response gaps occur in both Light and Cline, with provider/model/transport cause unknown; aggregate duration is not a causal harness speed estimate. A frozen counter mislabels successful string-valued finishes as failures; preserve it and separately derive zero explicit rejections from raw ledgers. An amendment also preserves nullable usage and correctly labels comparison predicates; no task, runtime, order or acceptance criterion changes. Manual review records signature/body editing retries and one optional compiler invocation preceding a final literal-only test edit; required invocations are fresh and final endpoint compilation passes. This small constructed project, dependent repeats and finite unblinded audits are not large-repository confirmation. Evidence and unchanged-patch replay: `docs/experiments/light-harness-selection-confirmation-f47-2026-10-18`.

24 F48: navigation and type diagnostics, implemented/local

Exact JS/TS declaration candidates now supplement bare-reference lexical fallback within 64 eligible files, 1 MiB read and four returned declarations. Scope, freshness, omissions and limits remain explicit; no identity is selected. This exposes and fixes a pre-existing alias-reuse defect for same-file homonyms: identity reuse requires a unique current declaration. Stale ambiguous aliases remain refused. Six constructed replays per arm require 15 versus 12 researcher navigation calls, but feedback grows from 21,552 to 25,302 bytes (17.4%). Timing overlaps local tests and proves no speed or model-token gain.

A separate bounded Vitest diagnostic distinguishes an explicit typecheck request, ignored source errors and reported type-error text while file selection remains unknown. Real Vitest 3.2.4 passes runtime despite TS2322 in an excluded new test; with explicit inclusion, it fails a typecheck suite while still printing "Type Errors no errors". Contradictory headings remain unknown and actual failures remain failed. No new command, scope admission or completion gate is added. Eighteen new controls and real runner receipts support this mechanism. The integrated runtime passes 1,020 tests with two skipped; runner 37/37, standalone payload 3/3, lint and line budgets pass. Earlier prototype/test-fixture failures are retained separately. Evidence: `docs/experiments/light-harness-navigation-type-evidence-f48-2026-10-18`.

25 F49: optional symbolic reasoning, local evidence

A fixed-seed constructed corpus has 18 supported scalar cases and eight refused controls. Exhaustive execution of 910 finite input assignments in actual transpiled JavaScript agrees with Z3: eight counterexamples replay and ten cases have no bounded counterexample. Unsupported cases remain unknown. A separate review compares 12 fixtures/85 inputs and hardens six prototype shape/binding/ assertion refusals before model exposure; these are new-prototype findings, not historical Light defects. The translator models only the isolated selected body, not module initialization, imported calls or all user requirements. Solver time is 386.8 ms versus 178.8 for enumeration with setup/VM work: single local observations establish no efficiency gain. Retain opt-in use and require real regression tests to assess witnesses in application context. Evidence: `docs/experiments/light-harness-symbolic-reasoning-f49-2026-10-18`; configuration and examples: `docs/light-symbolic-analysis.md`.

26 F50: optional symbolic exposure, new measurements

Three new constructed multi-file integer-policy requests cover credit, inclusive tiers and signed interval overlap, with two rotated repetitions and three arms. One frozen executable tree is used throughout: only the Light symbolic-exposure flag differs. Integrated Cline has ordinary command access to the same pinned Z3 binary and dynamic libraries. Prompt date, model/effort, dependencies, budgets, actual schemas and request limits are bound before calls. The optional schema/description and its cache effects are part of the treatment. Two additional integration controls pass at the actual offline Codex SDK schema boundary and in the standalone payload: ordinary tools remain identical and a real counterexample records no passing validation. No analyzer invocation or solver assertion is supplied in task instructions. The pure scalar initial source deliberately favors this prototype; valid solutions that introduce unsupported calls or loops remain valid and analyses remain unknown. Pristine, positive and semantic-mutation controls are completed before dispatch. Raw calls require separate relevance/domain review; a witness counts as exercised only with real-test evidence. Bounded absence grants no validation or user-acceptance admission.

All 18 official and selected outcomes pass; independent and strict reviewed counts differ. No `analyze` or direct solver command occurs in any arm. The actual Light schema grows from 10,473 to 12,278 bytes (1,805 bytes); this is tool availability, not observed symbolic reasoning.

Arm	Strict	Tokens	SDK USD	Seconds
Light, exposure off	5/6	442,954	.051857	443.4
Light, exposure on	6/6	521,734	.058953	459.2
Integrated Cline	4/6	546,832	.070161	336.4

All three primary failures concern direct negative-rate normalization in the exported `tierCost`. Each named consumer normalizes its rates correctly. The request names the consumer but not that helper's new behavior: the broader registered oracle is plausible, not an unambiguous user requirement. Preserve its failures and a uniform six-endpoint consumer-only sensitivity; do not use this difference alone as proof of superior task success. A separate post-hoc replay now executes 18,179 assertions per endpoint on the named count/quote functions over integers in [-3,3]: all six pass, with unchanged sources. This supplements the earlier static sensitivity and does not alter primary scores.

On five strict candidate/reference pairs, tokens/cost/duration increase 15.8/12.9/2.2%; four token losses and one win, leave-pair-out increases 5.8–32.4% and leave-task-out ranges -9.0% to +32.4%. All assigned tokens increase 17.8%. On four strict candidate/Cline pairs, tokens increase 4.3%, SDK cost falls 12.7%, and duration increases 60.6%; lower reported cost is not lower token consumption. Long response intervals in both Light arms have unknown

provider/model/transport causes. All 149 forwarded attempts have receipts; cache attribution per request still does not reconcile, and reported zero writes is not a measured absence. New SDK estimates total USD0.18097104; cumulative observations are USD8.3849118 plus USD3.64 historical reserves. Both the meaningful-use and efficiency gates fail. Leave default exposure off; retain the optional locally verified capability without claiming model gains.

A separate post-hoc diagnostic, selected after seven completed outcomes and registered with eleven available, applies 24 fixed queries uniformly to all 18 unchanged endpoints. Eighteen have no bounded counterexample, three remain unknown and three reproduce conditional witnesses in isolated actual JavaScript. Those witnesses concern the same ambiguous helper obligation, not new agent successes or indisputable user bugs. Sources and validation journals remain unchanged. This is capability/coverage evidence under supplied small domains, not confirmation or module-level proof. All 18 original patches also reproduce their five endpoint-check outcomes without repair. Evidence and reproduction: `docs/experiments/light-harness-symbolic-model-f50-2026-10-18`.

27 New local mechanisms: F51

Three limited changes address observed boundaries. First, inherited Node preloading observes actual supported Vitest invocations, including npm binary symlinks, without changing selection or wrapper control flow. Unique start/end records and fresh JSON reports bind the invocation, unchanged selected sources, Node version and manifest/lock identities; these are not full environment attestations. Collection failures with reported files remain failed even with zero executed cases; empty selection stays unknown. Twelve constructed old/current controls preserve masked failures, stale source, missing selection, unsupported reporters and a passing-runtime/failing-types distinction. None admits source verification or user acceptance.

Second, compact declarations and repeat-only unchanged-body guidance preserve explicit target choice. The known six-runs-per-arm replay retains twelve calls and reduces tool-feedback bytes from 25,302 to 24,060 (4.9%). This is still 11.6% above the older pre-F48 21,552-byte control; it is not a model-token or speed measurement. Third, three constructed formerly unsupported integer Math shapes now produce bounded absence of counterexamples. Shadowing, mutation, unsupported calls and unsafe domains remain unknown; assumptions include standard intrinsics and unmodeled imported effects. Discovery is opt-in and requires agent-supplied relevant properties and domains.

Final validation records 1,054 runtime passes, zero failures and two skips, 37 runner passes and four standalone-payload passes. Thirteen report tests include actual wrapped, masked, npm-bin, empty and setup-error invocations. Adverse development controls and subsequent fixes are retained separately: these are introduced prototype failures, not historical product defects. No timing claim is made for overlapping local runs. Evidence: `docs/experiments/light-harness-followup-f51-2026-10-18`.

28 Interrupted evidence and durable execution: F52/F54

F52 registered eighteen known-task discovery assignments. A cloud disconnect occurred after fourteen completed/manual-reviewed observations; assignment fifteen had started. The final driver state and subsequent execution are unknown. After restart the repository survived but the `/tmp` campaign did not. Fourteen hash-bound reviews and a partial audit packet survive; raw event/usage ledgers and endpoint patches are unavailable. These reviews cannot replace replayable audited successes. No final quality, resource or adoption comparison is claimed. All eighteen assignments remain explicit with unknown final metrics. The last verified F50 accounting is carried forward with USD13.50 of separate configured scheduling allowances for potentially charged F52 assignments; actual F52 spending stays unknown. This is neither imputed cost nor

an invoice upper bound. Frozen registration remains unchanged; F55 is a distinct known-task campaign, not replacement trials or fresh confirmation. Evidence: `docs/experiments/light-harness-symbolic-discovery-f52-2026-10-18`.

F54 changes only the research coordinator. New workspaces persist under `/workspace/blame-study-runs`, outside repository dependency ancestors. Every completed assignment receives an exclusive compressed checkpoint of raw files, hashes, observations and protocol identity, fsynced before workspace cleanup. Verified archives restore completed trials without model calls. A recorded exit and unchanged endpoint permit offline finalization; an uncertain started call blocks automatic retry/continuation and retains its unknown-cost reservation. Concurrent drivers are excluded. Cross-host/boot leases require inspection. This is local crash recovery, not a trusted signature or full-machine durability guarantee; regular remote commits provide another copy. Dedicated controls include a real SIGKILL, restoration, null preservation, corruption, protocol mismatch, credentials and concurrency. Evidence: `docs/experiments/light-harness-durability-f54-2026-10-18`.

29 Separate known-task discovery comparison: F55

F55 repeats the three known constructed requests in a distinct campaign after F52 evidence loss; missing F52 outcomes are never replaced or pooled. Two rotated repeats and eighteen frozen assignments compare identical Light bytes with body-read hints off/on and integrated Cline. Both Light schemas expose analysis and share the Math-capable engine. No property or forced use is supplied. All official, selected, independent and strict outcomes pass:

Arm	Strict	Tokens	SDK USD	Seconds
Light, hints off	6/6	599,297	.057277	278.1
Light, hints on	6/6	582,833	.064619	295.4
Integrated Cline	6/6	488,219	.072919	296.1

Sixteen hints produce no analysis calls; ordinary solver calls are also absent. Candidate/reference tokens decrease 2.7%, but SDK cost increases 12.8% and duration 6.2%. Leave-pair-out token changes span -13.9% to +10.4%, and leave-task-out -22.0% to +16.9%. Candidate/Cline tokens increase 19.4%, while SDK estimates decrease 11.4% and durations differ by -0.2%. Lower tokens and lower cost are distinct; neither adoption nor efficiency gate passes. Keep reasoning globally opt-in and retain the locally verified subset without a model-benefit claim.

All 157 forwarded attempts have receipts. Per-request cache attribution does not reconcile with SDK aggregates; reported zero writes remains a reporting limit. New estimates total USD0.1948144. Cumulative accounting retains USD8.5797262 observed and USD17.14 reserved, including unknown F52 consumption; this is not an invoice. One registered administrative pause after nine trials verifies actual recovery: their start/event/wire/patch/check/result hashes stay unchanged, and resume starts the next assignment with no repeated completed call. This changes overall wall time and may affect unobserved cache residency. Intermediate source, test-expectation and delivered-type errors remain in each trial's original cost/duration. All eighteen unchanged patches reproduce their five endpoint-check outcomes. Evidence: `docs/experiments/light-harness-symbolic-discovery-recovery-f55-2026-10-18`.

30 Larger known-project comparison and measurement correction: F53/F56

F53 registers three new constructed requests in previously studied Remeda (991 archived files, 208 production TypeScript files), two rotated repeats and eighteen assignments. These are small API changes, not a new-project or broad issue sample. Only executable `light-edit.js` and

`light-reference-search.js` differ between Light arms; identity safety, validation and all other bytes are common. Analysis is hidden in both schemas. All official and selected outcomes pass, but independent/audited counts differ:

Arm	Strict	Functional	Tokens	SDK USD	Agent s
Light reference	5/6	5/6	815,078	.073479	334.7
Light modified	6/6	6/6	1,499,485	.107510	457.6
Integrated Cline	2/6	6/6	1,936,119	.139177	465.7

The first reference suffix patch loses an existing index clamp: length two, count three returns one suffix item and retains one prefix item instead of all/empty. Its tests omit this finite oversized case. This is a model regression with insufficient cases, despite passing selected tests/types and honest scope reporting. Four Cline strict failures instead come solely from an updated preexisting Vitest results cache classified as a pinned dependency change. Preserve the registered scores. A separately registered post-hoc sensitivity applies the same narrow cache exception uniformly: conditional counts are 5/6, 6/6, 6/6. Old modified cache bytes and post-run manifest kinds were not retained, so this is conditional path/command-based diagnosis, not environment attestation or replacement primary scores.

Five strict modified/reference pairs increase tokens/cost/agent-time 90.8/44.6/31.0%. Two strict modified/Cline pairs increase 39.4/6.0/55.1%, below the four-pair threshold. Six conditional-cache/functional modified/Cline pairs decrease them 22.6/22.8/1.7%; leave-one-pair-out token changes span -34.0% to +15.1%, and omitting the slice task gives +4.2%. The last Cline slice trial uses 731,596 tokens after four delivered-type failures. No declaration-candidate or repeated-noop offer occurs in either Light arm; resource/quality differences cannot be credited to these unused mechanisms. Agents mostly use text/default searches, including a source-text query in symbol mode. Body replacements sometimes assume changed parameters/imports; validation then finds duplicate or missing names. The curried `splitAt` unknown-array inference limitation predates this study: agents initially ask stronger types of their new tests while the original API remains unchanged. These are distinct reasoning, navigation/edit-contract and validation-scheduling observations.

All 191 forwarded attempts have receipts; cache attribution still does not reconcile per request. New estimates total USD0.32016556; cumulative observations are USD8.89989176 plus USD17.14 retained reservations. SDK estimates exclude research/cloud compute pricing. Table durations are SDK agent time, not full campaign wall time. Separate timing ledgers record outer workers, endpoint grading and coordinator elapsed derived from root provenance/retained file metadata; the latter is not independent clock attestation. F55 includes its administrative pause. All eighteen F53 patches reproduce all five endpoint outcomes without repair. Evidence: `docs/experiments/light-harness-navigation-large-f53-2026-10-18`.

F56 corrects future research classification after the model cohorts are sealed. Only changes to existing regular `.vite/vitest/<40-hex>/results.json` become explicit `generatedModified` data; package changes, executable cache files and file/symlink transitions remain deviations. Future checkpoints also retain the post-run dependency manifest. Twenty coordinator/isolation/cache controls pass; a real isolated Vitest update is classified as generated and a subsequent package mutation is rejected, with the shared baseline unchanged. This is a measurement fix, not historical task recovery or full verification. Original registration revisions and primary results remain available. Evidence: `docs/experiments/light-harness-cache-integrity-f56-2026-10-18`.

31 Implemented feedback follow-up: F57

Three changes address the F53 observations without promoting navigation or test counts to validation. Ordinary/default/text searches and conservative declaration prefixes now expose bounded

exact syntax candidates, explicit symbol reads and compiler-reference routes. Original results, scope, homonyms and read-before-edit rules remain. A first known-Remeda control exposes the 64-file discovery limit: zero ordinary routes at baseline, two of four after initial exposure, four after visiting matching filenames first. The unchanged file/byte limits and incomplete labels remain; filenames order inspection and never establish binding identity.

Body-edit receipts show up to four observed input headers, each at most 512 Unicode code points, including unchanged and syntax-rejected bodies. Other explicit batch operations may change headers; no parameters or imports are automatically added. A real compiler control preserves the distinction between syntax admission and invalid duplicate/missing bindings. The identical local successful edit increases receipt bytes from 360 to 961; prevention by an agent is not established.

Repair advice retains literal cwd/task/argv identity and labels earlier-source failures as leads. It does not automatically replay raw commands or weaken checks/types. Finish prioritizes known failed selections and stops on a refreshed code failure. In a constructed replay it executes one check instead of three while the failure persists; after repair both versions execute all three required final-source invocations. Two saturated cross-file navigation controls use four scripted calls instead of six. These are local mechanism measurements, not model adoption, token savings or global latency. Baseline and targeted checks, introduced test/setup mistakes and corrected runs are archived separately. Evidence: `docs/experiments/light-harness-feedback-f57-2026-10-18`.

Final deduplicated runtime coverage is 1,068 passes, zero failures and two platform/pytest skips. The final full run has 1,063 passes and seven skips; five optional Vitest cases subsequently pass in a 13-case pinned-dependency rerun. These overlapping totals are not added. Runner and coordinator controls pass 37 and 20 cases respectively; runtime lint passes.

32 Isolated navigation diagnostic: F58

Eighteen new executions repeat the three known F53 requests, with two rotated repetitions. They are not fresh-task or new-project confirmation. Only three executable navigation modules differ; body-header and repair-order changes are absent in both Light arms. Model/effort, prompts, schemas, pinned environments, budgets and all five endpoint checks are frozen before outcomes. A pre-dispatch binding guard detects a generated Vitest cache changed by the last local tests: zero model starts occur. Original bindings/refusal and a common cache-only amendment are retained before the first model call; no trial is retried.

All official and selected outcomes pass. Strict results preserve one real reference regression: removing index clamps loses existing behavior for finite counts above array length; its new tests omit that case. No source is repaired.

Arm	Strict	Tokens	SDK USD	Agent s
Light reference	5/6	1,043,147	.090954	413.0
Light navigation	6/6	921,018	.086742	370.5
Integrated Cline	6/6	1,980,063	.136406	512.9

Five strict navigation/reference pairs reduce tokens/cost/time by 19.4/13.3/13.7%. Omitting individual pairs retains token reductions of 5.3–29.9%, but omitting clamp reverses the token result to +20.3%. Six strict navigation/Cline pairs reduce them by 53.5/36.4/27.8%; token reductions remain 30.1–58.3% when omitting one pair and 25.1–64.6% when omitting one task. This is a favorable efficiency signal within this known panel, not a general advantage or attribution to the new navigation.

Nine new declaration offers have zero direct symbol/reference follow-ups; agents often read whole small API files. Reference has one existing declaration offer, also unfollowed, and one repeated-noop offer followed by a read. Implicit information use cannot be attributed from the trace. The registered quality and resource gates pass, but the mechanism gate fails. Retain

the locally verified mechanisms; do not claim incremental model benefit for navigation. Header prevention and repair-order effects on agents remain unmeasured here.

The last Cline slice trial consumes 829,390 tokens after three new-test type failures. Six already-green checks are replayed during unsuccessful repairs; the three final-source required checks remain necessary. The stronger curried/pipe assertions exceed preexisting outer-generic inference; public types and original tests are preserved. Another correct candidate's mutable-limit test changes only bound order, leaving the normalized range unchanged: that assertion would not detect snapshotting. Its source reads per invocation and the independent oracle changes the actual range. Passing authored tests and satisfying user behavior remain distinct. Formatter availability/policy friction is retained.

All 186 forwarded requests have receipts; cache attribution per request remains unreconciled and reported zero writes is not measured absence. New SDK estimates total USD0.31410156; cumulative observed/reserved accounting is USD9.21399332/USD17.14, retaining unknown F52 consumption. SDK times exclude setup, grading and research; separate timing ledgers bound those claims. Original execution sources are archived. A positional audit-wrapper error is fixed after all model exits without changing the frozen trials or primary scores. Immutable raw reviews, accounting, verification and replay evidence: `docs/experiments/light-harness-navigation-feedback-f58-2026-10-18`.

33 Feedback ablations and fresh-project confirmation: F59–F61

All three panels are selected before new model calls, with three requests, two rotated repetitions and reference/candidate/integrated-Cline arms each. F59 changes failed-check advice and finish ordering (two executable modules); F60 changes body-edit header receipts (one). Other current mechanisms stay common within each panel. These repeat known Remeda requests with natural tool choice and no seeded history. F61 compares the complete current feedback bundle with executable baseline `b70c21fa` on new cross-module TS-Pattern `d68fbcf1` requests. Its eighteen production TypeScript modules do not establish large-project representativeness. References differ between panels; keep outcomes and resources separate.

Exact prompts, schemas, model/effort (gpt-5.6-luna/high), dependencies and budgets precede each dispatch. Five endpoint controls separate selected upstream regressions, finite behavior/public types, source compilation, new runtime tests and delivered-test compilation. Original checks/configuration stay protected. Full structural requirements and truthful final-source reporting are manually audited beyond those controls. No failed patch is repaired or retried. All 54 original patches reproduce their five endpoint outcomes independently, including retained failures.

Arm	Finite	Audited	Tokens	SDK USD	Agent s
F59 reference	6/6	6/6	1,348,074	.103946	460.5
F59 repair	6/6	6/6	950,319	.087575	400.9
F59 Cline	6/6	5/6	1,619,425	.126532	424.7
F60 reference	6/6	6/6	1,161,629	.101043	430.9
F60 headers	4/6	3/6	1,205,733	.098848	423.5
F60 Cline	6/6	6/6	1,498,379	.127101	599.4
F61 reference	6/6	5/6	2,323,918	.153025	772.3
F61 bundle	6/6	6/6	1,786,607	.144607	963.5
F61 Cline	6/6	6/6	3,898,754	.235506	756.4

Runtime completion is 6/6 in every arm of each panel. Finite endpoint success and strict audit are distinct. Light's invocation-level passing tests still leave source verification *unverified*; independent audit supplies different bounded evidence, not runtime admission or universal user acceptance.

Failed-check priority. F59 has six strict candidate/reference pairs: tokens/cost/time fall 29.5/15.7/12.9%. Omitting clamp retains 14.1% token savings; slice alone rises 20.0%. Five strict candidate/Cline pairs fall 33.6/25.5/2.6%. However, four advisory receipts have zero direct priority follow-ups; no automatic finish refresh is needed after the final required checks already pass. The mechanism gate fails. Retain the local scheduling control, with no demonstrated incremental model benefit. Three same-source duplicate required successes in one Cline trial are separate from greens before a renewed failing repair. Formatting/test edits invalidate previous final-source checks. The final Cline slice patch passes every endpoint but adds a test newline after validation: a strict freshness failure, not a functional regression.

Body-edit headers. F60 forwards three complete declaration contexts (1,896 unique JSON bytes); all three body edits already have correct bindings. No exposed erroneous-body recovery is observed. A reference body edit causes duplicate/missing bindings and is repaired by whole-declaration replacement. Both candidate suffix patches remove consumer index clamps while leaving the shared helper unbounded: length two/count three fails despite passing authored tests. Neither uses body edits or headers, so these adverse outcomes do not identify a receipt effect. A correct reference keeps consumer clamps with an unbounded helper; helper boundedness alone is not a user obligation. The first candidate clamp is correct with fresh required checks but falsely reports an optional test compiler as run after its last formatting edit. Strict reporting and endpoint correctness remain separate. All-assignment candidate/reference tokens rise 3.8% with worse quality. The three surviving strict pairs (24.7% fewer tokens than reference, 38.3% fewer than Cline) miss the four-pair minimum and exclude all suffix assignments. Quality/resource gates fail. Keep the bounded receipt as a diagnostic aid; no success/token benefit or payload expansion is justified by this panel.

Fresh-project confirmation. F61's eighteen production implementations meet the inspected structural and behavioral requests. The first reference own-key trial delivers a new test whose broad record pattern fails TS2345; source compilation excludes it. The agent omits the optional test compiler and honestly reports its three required checks. This accounts for the strict 5/6 versus 6/6, separately from production success. Against Cline's equal audited count, current Light saves 54.2% tokens and 38.6% SDK cost, with 27.4% longer agent time. Token savings survive all omitted pairs (48.3–60.8%) and tasks (38.5–60.9%). Against reference, all-assignment tokens/cost/time change by $-23.1/-5.5/+24.8\%$, with unequal audited quality; five strict pairs change by $-20.4/-0.27/+48.1\%$. Quality and exposure gates pass, duration/resource gate fails. The machine outcome flag is true because strict count increases by one with observed uptake; it does not establish efficiency, mediation or a stable success-rate difference.

Eleven candidate declaration offers yield one symbol-read follow-up. Twenty repair receipts yield five literal next-validation matches but only three distinct later invocations; repeated advice is not multiple prevented errors. No arm uses body replacement or receives headers. Some authored tests use separate clauses or discriminate before selecting, recreate patterns in freshness cases, or return an empty fallback that can mask nonmatching. Source review and independent finite controls supply the missing evidence; authored test names alone do not. Rejected scope errors remain atomic. Retain bounded diagnostics, abandon a general efficiency claim, and instrument latency before attributing the additional time or changing concurrency. SDK request/tool sums are not network phases or a provider-causality estimate.

Measurement and confirmation boundaries. The benchmark recorder masks every field named `headers`, including function declarations. Actual SDK tool responses retain them. Before F60, an identical passive preload in all arms measures hashes/counts/sizes from forwarded tool messages and joins request hashes without archiving declaration, conversation or credential text. Masked event bytes are not model feedback; context JSON sizes are not added billed tokens.

F59's old secondary byte counter is explicitly qualified. Cache attribution and invoices remain uncertain; new F59/F60/F61 SDK estimates are USD0.31805308/USD0.32699292/USD0.53313776 (USD1.17818376 total), with cumulative observed USD10.39217708 and retained USD17.14 reservations for unknown historical consumption. SDK times exclude setup, grading and research; a 258.3-second Cline trial remains. Detailed timing ledgers bound those claims.

Before any F61 call, static oracle review permits omitted selections on a failed matcher, as allowed by upstream `MatchResult`. A valid omitted-selection control passes all five checks; a false-only leaked-value mutant is rejected. Successful union initialization remains required. Original unbound registration and controls are retained; tasks, runtime bytes, rotation and budgets stay unchanged. F61 uses selected original Jest-style tests through the pinned Vitest study adapter, with separate source/new-test compilers. It is not the complete native upstream suite. Protocols, source archives, raw checkpoints, reviews, measurement qualifications and replays: `tools/terminal_bench/feedback-ablation-study` and `docs/experiments/light-harness-feedback-f59-2026-10-18`, `docs/experiments/light-harness-feedback-f60-2026-10-18`, `docs/experiments/light-harness-feedback-f61-2026-10-18`.

34 Latency, boundary sensitivity and final evidence: F62

Three follow-ups are selected at `c1e6d5f9` before implementation/model outcomes. The production change is limited to three executable modules: post-change invalidation feedback, current/historical check presentation and the completion report. Exact cwd/argv/source identities remain separate; oversized identities are omitted with counts. Current rows expose recorded time and paths, with bounded truncation. Successful ordinary commands are not displayed as admitted checks. Source-changing commands and completed handles share the observation; pending handles and noops do not. Required-check/admission decision algorithms are unchanged. This feedback neither rechecks environment nor proves invocation start time, arbitrary prose or user acceptance.

Local diagnosis and controls, not new model results. F61 tool intervals account for 100.9/86.5/93.0 seconds in reference/current/Cline; total SDK times are 772.3/963.5/756.4 seconds. The current/reference increase is 191.3 seconds, comprising 14.3 fewer seconds in tools and 205.6 more outside their recorded spans. Those older traces cannot separate gateway, transport, generation and client work. New common demand-driven HTTP observation records headers, first body read and EOF/cancellation/error without eager consumption or archiving bodies/credentials. Interval unions/intersections retain overlap; consumed-stream time is not pure server inference or network latency.

All six archived F60 suffix test sets pass both faulty candidate implementations (12/12 missed defects); six self controls reproduce their original passes. Two of six F61 alternative suites pass pristine faulty production; four fail active cases. These unchanged-test sensitivity diagnostics never repair patches or rescore historical trials. The original F60 oracle already rejected its defects. A new common finite checker adds length-relative near-threshold probes through length 24 and changing-bound reuse; positive controls pass, and an unclamped-index mutant fails the new-only probes. Full runtime validation passes 1,068 tests with seven declared skips; thirteen separate invocation-report checks execute the five skipped real-Vitest wrapper cases (eight overlap). Runner checks pass 37/37 and timing controls 12/12, including actual SDK/common-preload metadata joins. An unavailable historical temporary solver path and an unbound snapshot taken during compilation are corrected before any model dispatch.

New model measurements. All eighteen registered assignments complete: three known Remeda diagnostics, two rotations and current-reference/provenance/integrated-Cline arms. The reference is the sealed F61 current executable, not its older reference. Exactly three

executable modules differ between Light arms; observation and boundary checks are common. Source `f74da68b` and exact bindings `d5f42dd2` precede calls. Prompts, schemas, model/effort, environment and budgets remain fixed; no forced history, null-opportunity replacement, retry, endpoint repair or recovery of a solution occurs. Runtime completion is an agent signal, not independent acceptance.

Arm	Complete	Oracle	Strict	Tokens	SDK USD	Seconds
F61 current reference	6/6	5/6	4/6	1,393,915	.107108	436.3
Provenance candidate	6/6	6/6	5/6	963,560	.089259	423.6
Integrated Cline	6/6	6/6	6/6	1,311,599	.113258	388.9

The reference suffix failure repeats the unclamped length-two/count-three model defect despite passing authored tests. Its second clamp trial has correct final code and required invocations, but falsely calls an earlier optional compiler and whitespace check final-source evidence after a formatting edit. Candidate splitAt repeat two passes production and 27 runtime tests, but its new test fails TS2322: a numeric-array annotation assumes inference absent from the preserved curried API. A post-hoc unchanged-test probe reproduces the identical compiler error on pristine and patched production, isolating the test/API assumption from a new production-body regression. Its `test:study:delivered - types` command actually repeats runtime: the package script already selects that mode. The final report honestly excludes test compilation, but the delivered test still fails strict acceptance. Every patch/command/delivery is manually audited; all eighteen unchanged endpoint replays reproduce their five recorded check outcomes.

All six candidate trials edit before project validation: zero invalidation offers, forwarded receipts or confirmed same-source reruns. The registered mechanism gate fails. Compared with Cline, all assigned candidate trials use 26.5% fewer tokens and 21.2% lower SDK estimates, but 8.9% more time and lower strict quality. This is not efficiency at quality parity. On five jointly strict-successful Cline pairs, savings are 24.9% tokens/20.4% cost with 9.2% more time; leave-one-pair-out token savings range 12.5–33.0%. The failed assignment remains in primary totals. Only three jointly strict reference pairs remain, below the registered minimum four. Quality versus reference passes; resource and model-benefit gates fail. These dependent repeats do not establish causal mediation or representative gains.

New time attribution and decision. Recorded tool/HTTP/outside spans total 88.5/333.6/14.2 seconds for reference, 71.7/339.4/12.5 for candidate and 47.4/328.3/13.2 for Cline. All 178 streams complete with no interval anomalies or tool/HTTP overlap. Candidate/Cline's extra 34.6 seconds comprise 24.3 extra tool seconds, 11.1 extra HTTP seconds and 0.7 fewer outside both. Candidate makes 59 calls versus Cline's 46; its header/body-tail spans are 67.1/272.0 seconds versus 44.9/283.3. More round trips and tool work are measurable costs, without proving generation/network causality or explaining historical F61 from absent HTTP data. No scheduled development build/test/compiler or solver overlaps live workers; provider/host exclusivity remains unknown.

All input/output token and SDK-cost sums reconcile; reported aggregate cache reads are 1,149,056/760,576/1,036,928 and already included in input. Per-request cache attribution remains unreconciled; null first receipts stay null. There are 178 wire attempts and usage observations, zero timeouts and no new missing-consumption reserve. New SDK estimates total USD0.309625, bringing observed cumulative estimates to USD10.70180208; retained unknown reservations USD17.14 give USD27.84180208 accounted. Billing, hidden retries and EUR conversion remain unknown.

Retain the history/admission correction, observer and finite controls for tested accuracy and diagnostic value; no model efficiency benefit from the unexposed receipt is demonstrated. Next isolate an identical post-validation edit handoff, explicit optional-check claims and wrapper argument semantics, then test round-trip reduction on preselected larger cross-file tasks. Keep these diagnostics separate from confirmation. Artifacts, audits, raw traces and frozen coordinator sources: `docs/experiments/light-harness-feedback-f62-2026-10-18`.

35 Controlled validation recovery and RxJS confirmation: F63–F64

Implemented and locally tested. Runtime-test results now expose a bounded scope diagnostic when TypeScript tests change or extra arguments are forwarded. It shows the declared script, argv, changed test paths and real catalog typecheck selections. Static test compilation remains unestablished; wrapper argument names and runtime cases prove no compiler mode or file membership. Commands, schemas, required checks and admission stay unchanged. Source `79cc3399` passes 1,071 runtime tests with seven declared skips; three scope controls, 37 runner and 12 timing controls pass. Supplemental wrapper checks execute five real-Vitest skips (eight cases overlap the full suite). The new integration assertion fails against sealed F62 current as expected.

Selection and experimental boundaries. Selection `66528075` registers two separate eighteen-assignment panels. F63 provides an identical controlled post-validation edit handoff, with or without the real serialized invalidation field; both Light executables are identical. Source, test-only and wrapper-mode cases preserve source identity including Git HEAD. Worker journals start empty: the user handoff restores no admitted check and is measured separately from live tool receipts. F64 isolates the scope diagnostic against sealed F62 current on three new RxJS operator requests, requiring implementation, both public barrels and delivered tests. The complete fixed upstream checkout contains 1,288 files and 251 library TypeScript files; it is larger than F61, but these constructed features do not establish issue representativeness or general large-project savings. Selected original Mocha cases use a common callback bridge and ES2015 class-field semantics with explicit adapter controls. Positive controls pass all five endpoints; state-sharing and test-type mutants fail. Completed panel artifacts: `docs/experiments/light-harness-validation-recovery-f63-2026-10-19` and `docs/experiments/light-harness-rxjs-confirmation-f64-2026-10-19`.

F63 new model measurements and adverse result. Each arm passes official completion and strict audit in 6/6 assignments; all 18 unchanged patches reproduce their five endpoint outcomes. Candidate versus reference uses 10.2% more input-plus-output tokens, 8.0% more SDK estimated cost and 15.5% more SDK duration. Versus Cline it uses 4.2% more tokens, 33.0% less estimated cost and 26.8% less duration. All six candidate user handoffs are observably forwarded, but the resource gate fails. Abandon an efficiency claim for this reminder on this panel; retain the local source-invalidation contract. Fresh journals explain empty `@checks` reads; the reference also reads that ledger. An artificial handoff cannot establish natural opportunity rates or same-session mediation. The panel costs USD0.15046948 in SDK estimates with no new unreported-request reserves; aggregate cache is included in input tokens, not added again, and per-request cache reconciliation remains incomplete.

Measurement limits and prospective correction. After the first F63 SDK completion, collection failed on a tracked dependency symlink replaced by the isolated directory before the model. The uniform grader now excludes only that infrastructure anchor from source diffs, retaining raw diffs and full dependency manifests. Original bindings are archived; no trial is retried or source repaired. That trial's OS exit, outer duration and atomic source capture remain unknown; SDK usage and duration are available. Separately, actual installed-provider local serialization controls reveal an observer blind spot for Responses `input_text` and custom outputs. F63 live-receipt zeros (and older zeros under this observer) do not prove no exposure; historical bodies cannot be reconstructed. Independent explicit-user-handoff exposure is valid. Before F64 dispatch, typed-wrapper metadata is extended and tested on actual provider JSON, content and cached-JSON

serialization without network calls. User/source JSON stays excluded and requests remain unchanged. This measurement correction changes neither runtime treatment nor historical scores.

F64 new measurements: favorable harness comparison, unproved treatment. Official completion and source audit are 6/6 for all arms; strict delivery audit is reference 5/6, candidate 6/6 and Cline 6/6. The sole strict miss places a new test under `src/internal/operators` rather than the requested `src/*.test.ts`; it runs and compiles. Treating that layout as equivalent gives 6/6 throughout. This difference establishes no behavioral or reasoning gain. All eighteen unchanged patches reproduce their five endpoints, bringing the two panels to 36 reproducible assignments without researcher solution repair.

Against Cline on six strict pairs, candidate uses 17.4% fewer tokens, 13.9% less SDK estimated cost and 1.0% less SDK duration. Token savings survive omitting any pair (11.9–23.3%) or task (14.1–18.6%). Against reference on all six source-correct pairs it uses 1.8% more tokens, 0.7% more estimated cost and 5.4% less duration; on five strict pairs, tokens rise 4.2%. The marginal resource gate fails and omission changes that token direction. Candidate makes 95 forwarded requests versus Cline's 73; observed tool time is 181.5 versus 130.5 seconds and request spans 697.6 versus 760.8 seconds. Queue, thinking and transport are not separately identified. These constructed features support a small-panel harness comparison, not general issue resolution or a diagnostic-effect claim.

The treatment's 32 scope offers are archived in tool ledgers, but no typed scope receipt is recognized in forwarded contexts. Twenty-one common invalidation receipts are detected. Full SDK-loop visibility of scope feedback remains unestablished despite the expanded provider controls; the mechanism gate fails. Retain local advisory scope behavior, abandon a claimed marginal model gain, and resolve actual message-path visibility before another causal comparison.

F64 verification amendment and accounting. The frozen checker rejects unequal prompt contracts: generated Git HEAD differs because study adapters contain checkout paths. After all model exits, a separate checker verifies each archived input commit and workspace, then normalizes only those metadata fields; all six Light templates match. Raw prompts and request hashes remain unequal and archived. This incidental context limits isolation; no dispatch, score, source or consumption is rewritten. Positive and negative normalization controls pass; the original failed checker log and both versions are retained. F64 costs USD0.54828440, the two panels USD0.69875388 in SDK estimates, with no new unreported-request reserves. Cumulative observed estimates are USD11.40055596 plus unchanged historical reserves USD17.14. Aggregate cache is included in input; per-request attribution and invoices remain incomplete.

36 Feedback delivery and isolated follow-ups: F65–F67

F65 diagnosis, implementation and local evidence. All 32 F64 scope offers are top-level fields in saved tool results; nested placement was not their cause. The presenter emits a metadata JSON line and fenced diagnostic text. Whole-string JSON parsing missed 32/32 locally replayed shapes; the corrected observer matches all 32 exact hashes. Original provider bodies remain unavailable: F64 exposure and scores are unchanged. Full SDK0.2.0 OpenAI-compatible loopback controls additionally expose real 8,000-character clipping before custom projection. Across projection on/off and normal/long output, exact receipts survive 2/4 raw cases and 4/4 corrected cases. Large `run_task` results now separate structured metadata from diagnostic text in typed SDK parts; the projector and app decode this envelope. Oversized metadata can still be unavailable. Long-output forwarding grows from about 8,506 to 13,772 characters; preserved evidence is not demonstrated token efficiency. No compiler scope, case count or admission claim is strengthened by transport.

The final pre-migration prototype suite passes 1,188 runtime tests with 13 skips; app tests pass 396/396 and runner 37/37. A stale preview-contract build caused one initial failure; rebuilding dependencies resolves it without test relaxation. Nine final receipt/source/native/evidence controls pass. A post-dispatch negative control extends detached-JSON rejection to typed `input_text` envelopes; it fails before the one-line observer correction. Raw receipts are untouched. None of 134 inspected command diagnostics contains the scope literal needed for that false positive. Independently, all 31 unique live scope offers match an exact hash and call identity at the fetch boundary. This proves neither attention nor use; no historical request reconstruction or score correction is claimed.

Frozen protocol and interruption. Selection `71905418` precedes outcomes. Each panel selects two new requests, two repeats and three arms: F66 tests individual/grouped reads/Cline on the same complete pinned RxJS project reused from F64; F67 tests literal compiler priority off/on/Cline on new small fixtures with a seeded error. Both Light arms share F65. SDK0.2.0 native tools have no historical research search patch. The common runtime is frozen at `80e3e881`; schemas, prompts, SDK/core/provider bytes, Git input identities, absolute workspace, copied dependencies, offline npm caches, model `openai-codex/gpt-5.6-luna/high` and budgets are bound before dispatch. Analysis/hints are disabled. Limits are 35 iterations, 600 seconds, 8,192 output tokens and USD1 SDK cost per trial, with an explicit scheduling reserve. Four required final invocations and five independent immutable endpoints remain. Four positive controls pass all endpoints; four behavioral mutants are rejected.

The original coordinator receives SIGKILL during assignment nine after eight checkpoints. Kernel evidence confirms the signal, not its sender or the model's exit. That trial's final source, aggregate usage and duration remain unknown; eight usage observations and the partial patch are retained. A pre-outcome continuation binds the same executable bytes and the original fifteen remaining assignments, one new trial per process. There is no retry, replacement or solution recovery. All 24 assignments are accounted for; 23 complete. All 23 unchanged model patches reconstruct exact final-source hashes from the published original-input Git bundle and reproduce all five endpoint outcomes. The incomplete trial stays in assigned denominators and is excluded explicitly from matched success pairs.

Arm	Complete	Source	Strict a/m	Tokens	SDK USD	Seconds
F66 reference	3/4	3+?	2/2	unknown	unknown	unknown
F66 batch prototype	4/4	4/4	2/2	1,980,624	.117260	448.0
F66 integrated Cline	4/4	4/4	1/3	2,138,292	.146845	498.7
F67 reference	4/4	4/4	4/4	255,241	.036620	421.7
F67 compiler priority	4/4	4/4	4/4	267,003	.033777	156.7
F67 integrated Cline	4/4	4/4	2/4	248,900	.042747	222.7

Strict a/m gives automatic/manual-audit counts, each out of four. Source 3+? means three passing and one unknown. The conservative frozen native observer excludes loops, `cd` wrappers and edits followed by checks within one call. Post-hoc inspection accepts five such cases from exact child-status and source-order evidence; their original automatic scores remain intact. A shell's final zero alone is never used. Strict layout still fails for the reference's first map test, both batch take tests and Cline's second take test: tests work but are below a subdirectory instead of the requested `src/*.test.ts`. These are delivery failures, not demonstrated production-behavior regressions. All terminated patches pass finite independent behavior, public-type, regression and delivered-test checks.

F66 adverse result and decision. The bounded prototype preserves per-target status/aliases, signatures, aggregate 6,000-character limits and rollback of omitted observations. However, none of four models uses it. The reference already issues multiple reads in a single model iteration (seven

such iterations in its first map trial); a new batch syntax does not necessarily remove HTTP round trips. The prototype makes 63 requests versus Cline's 47. On all four source-correct pairs it uses 7.4% fewer tokens and 20.1% lower SDK estimates, at lower strict quality. On only two jointly strict Cline pairs, tokens instead rise 22.0% and duration 14.3%; no jointly strict reference pair remains. Reject a marginal efficiency claim and remove the unused production prototype. Its tested source and compiled executable remain archived for replay.

F67 implemented behavior, measurements and decision. Recognized literal `tsc -noEmit` and local TypeScript Node-entry commands, optionally with a literal project path, can lead repair and refresh before green project checks. Evidence stays invocation-only; every final required selection is retained. Compound/expanded/emit-enabled commands are excluded. Local controls verify ordering, exclusion and stopping before green checks when repair still fails. The frozen raw next-command counter includes inspections; a separate event-order audit excludes those and counts one opportunity per actual failed compiler. The exact compiler finishes before required checks start in 4/4 corrected runs versus 0/4 reference. Success is equal, not improved.

At that bounded quality, candidate/reference tokens rise 4.6%, SDK estimates fall 7.8% and raw duration falls 62.8%; both make 32 requests. The reference's first quota stream alone lasts 237.1 seconds before any tool: it cannot be evidence of later repair advice. Omitting that pair leaves 11.0% lower duration. Omitting one task changes token effects from −14.7% to +34.0%, and SDK cost from −16.5% to +3.0%. Versus Cline, tokens rise 7.3%, SDK estimates fall 21.0% and duration falls 29.6%. Retain accurate compiler priority; task-success improvement and robust general efficiency remain unestablished. Repeated repair advice and navigation turns are concrete follow-up candidates, not proven causes of token differences.

F65–67 accounting and integration boundary. All 23 completed token/cost aggregates reconcile with their usage observations; per-request cache attribution remains incomplete. Cache is an input subset, never added to input-plus-output. Completed SDK estimates total USD0.46597056; partial observed cost USD0.01469516 gives USD0.48066572 observed, with USD1.50 new unknown-trial scheduling reserve, not an invoice bound. Historical observed estimates become USD11.88122168 with USD18.64 retained reservations. Complete global consumption, hidden retries, invoices and EUR conversion remain unknown. Passive HTTP/tool interval unions and source-observer overhead are retained; queue, reasoning, generation and transport causes are not separately identified. The new USD30 scheduling stop is separate from old cohorts. At F65–67 completion, the later dependency/runtime migration is integrated and verified separately; that suite passes 1,189 runtime tests with 13 skips, 396 app tests, 37 runner tests and nine research controls; app typecheck, affected lint and line budgets pass. Two new fixture failures after migration came from a missing CLI in the compatibility package; fixtures now use the existing real-compiler setup and assert TS2322. No production behavior or benchmark result was changed by that fixture repair. The full SDK loopback contrast remains 2/4 versus 4/4 on the integrated stack. Model results remain on the frozen pre-migration executable, not a new-stack comparison. Source, controls, exact patches, byte bindings, negative evidence and replay inputs: `docs/experiments/light-harness-feedback-delivery-f65-2026-10-27`.

37 F68–70: context economy and bounded declaration discovery

Implemented behavior and local controls. Repair-plan compaction is a pure model-context projection: only an identical advisory plan with a full copy still visible in the same transcript receives a short reference. New failures, source/requirement changes and explicit `read_code @checks` reads expose full advice again. Raw journals and validation admission are unchanged. An actual SDK loopback control forwards 907 characters versus 343 for the duplicate, preserves the previously forwarded prefix, and retains the failed outcome. Compaction is now opt-in because

a useful general efficiency effect has not been demonstrated. Literal filename matching fixes the recorded `test.ts` query that previously suggested `combineLatest.ts`; ordinary word and lexical searches remain available.

Prospective measurements, kept separate from historical campaigns. Tasks, arms, repetitions, settings and budgets were registered before dispatch. F68 reuses two known diagnostic tasks; F69 uses two historical TanStack Query issues new to this study, from full monorepo checkouts. The mutation task's upstream fix touches one production file; the filtering task is a core/React subset of an eight-adapter change. Each task has two fresh draws per arm. The common executable uses SDK 0.2.0, `gpt-5.6-luna`/high, 35 iterations, 600 seconds, 8,192 output tokens and a USD1 per-session SDK-estimate cap; input sources, schemas, prompts, major SDK/provider bytes, dependencies and graders are bound. Light treatments change one feature at a time. Compiler priority is common; optional symbolic analysis is disabled. These results are not pooled with F65–67.

Panel	Arm	Source/audit	Tokens	SDK USD	Seconds	Requests
F68	reference	4/4	357,187	.036775	187.2	39
F68	compact	4/4	323,437	.037274	190.5	36
F68	Cline	4/4	312,513	.043481	240.0	46
F69	reference	3/2	2,627,325	.143048	760.0	88
F69	compact	3/2	2,531,504	.140371	687.5	86
F69	filename	2/2	2,268,648	.134467	640.1	76
F69	Cline	3/3	2,262,157	.154951	520.7	46

Source/audit are counts out of four, not a ratio: source passes the finite independent endpoints and immutable-input guards; audit additionally requires runtime completion, delivered tests and observed required invocations on final source. F68 has four of each per arm. F69 callback patches all pass; three filter patches pass the independent source endpoints, but only Cline's second filter run completes the full audit. Failed assignments remain in resource totals.

Adverse evidence and causal limits. F68 compaction/reference tokens fall 9.4%, but SDK estimates rise 1.4% and duration 1.8%. Omitting the clamp task instead raises tokens 37.4%; omitting quota lowers them 36.9%. Against Cline, tokens rise 3.5%, while SDK estimates fall 14.3% and duration 20.6%. Only one of eight compact-arm draws across F68–69 actually emits a short reference. These resource differences cannot be attributed generally to compaction. Filename matching repairs a navigation defect; its small live panel does not establish a task-success or efficiency improvement.

The filter failures expose a separate reasoning/testing gap: a generic constrained by `Omit<...,'fetchStatus'>` can still accept that field alongside an allowed field. Negative tests containing only the forbidden field pass for an unrelated weak-type reason. Unchanged upstream tests and independent mixed-field probes reject these patches despite green authored checks. Two other Light patches are source-correct at the finite endpoints but lack the frozen required-execution proofs; one is stopped by the SDK duplicate-call guard while polling a managed check. Post-hoc inspection corrects the second filter reference diagnosis: its last poll shows "exited" and 631 passed cases, but neither an exit code nor a finalized result. The hash-bound original journal records exit zero 225 ms later, with unsupported test reporting still unknown. This exposes a process/result-publication interval, not proof that the test never ran or solely a model omission. The primary tool-receipt score stays unchanged; journal diagnostics are archived separately. Execution, passing, formally admitted verification and user acceptance remain distinct. Missing native obligation records are unavailable, not failed checks; unsupported Vitest 4/5 reporting does not establish project verification.

Across the twenty Light draws, all 32 unique offered test-scope receipts match actual fetch-boundary payloads; this establishes forwarding, not remote ingestion or model attention. Available

grouped reads are used zero times. The common compiler-priority advice is followed in all eight observed literal-compiler recovery opportunities, before required validation starts; it is not the ablation in this campaign, so no new causal success claim follows.

Second cycle, registered before its model phase. Recorded declaration-prefix searches exhaust the 1 MiB syntax-discovery budget on tests before reaching `types.ts`. The targeted follow-up preserves exact filename affinity, then visits other source before conventional test paths. Limits, freshness rechecks, explicit declaration choice and lack of read/edit coverage remain intact. Local byte/file-budget controls and actual SDK transport checks pass. Four real-query replays match the sealed reference with the flag off; with it on, two previously missed declarations become available without changing primary lexical/symbol results. Current validation passes 1,199 runtime tests with 13 skips, 37 runner tests, eight research controls, affected lint and line budgets. The twelve-draw F70 comparison uses the same two known diagnostic issues and a shared clarification of generic API compatibility, with a separate compiler probe. It is a new specification/audit cohort, not fresh confirmation.

F70 arm	Source/audit	Tokens	SDK USD	Seconds	Requests
Reference	3/2	2,883,054	.155681	820.2	90
Source order	3/2	2,359,983	.146966	817.5	80
Cline	4/4	3,983,456	.197298	758.0	71

All four Light filter draws hit the SDK's five-identical-call guard while polling an active regression command; two patches subsequently pass all independent endpoints and two fail compilation of their own added tests. This is distinct from exhausting 35 iterations. Both callback repetitions pass in every arm. On these two jointly audited callback pairs, source order/reference uses 24.9% fewer tokens, but leave-pair-out effects range from -59.0% to +25.0%. Against Cline on those pairs, tokens fall 44.3%, SDK estimates 31.2%, and duration rises 5.5%; the filter failures are excluded only from this paired sensitivity, never from the primary totals. Eight offered declarations are subsequently read in the candidate's two filter draws, versus none in its callback draws. Transport and use are demonstrated, but no general task-success or causal efficiency gain is established. Retain the bounded navigation correction for local declaration recovery; prioritize process observation over further navigation promotion. F70 adds USD0.49994476 in complete SDK estimates.

Accounting and reproducibility boundary. All 28 original task assignments terminate and are retained, with USD0.69036568 in reported SDK estimates. A transport interruption ends after assignment 24; its recorded exit/checkpoint are recovered without retry. The last four assignments run after a host restart, with the frozen bindings still enforced. SDK time excludes setup and independent grading; passive observer overhead is reported, not subtracted. Per-request cache/reasoning remains incomplete; cache is an input subset and SDK estimates are not invoices or EUR conversions. The original Vitest 5 image includes hash-bound preflight caches. No cache-solution retrieval is observed, but these images are not claimed to be clean; the follow-up uses cache-free images and private grader copies. Grader-added cache files are removed only to restore the exact original image before the next assignment. Original patches, end-of-run permission metadata, test diagnostics, executable bindings, input commits and cache bytes are archived for independent replay; patches are never repaired. Evidence and protocols: `docs/experiments/light-harness-context-economy-f68-2026-10-27`; `docs/experiments/light-harness-declaration-order-f70-2026-10-27`.

38 F71–72: managed-process observation

F71 addresses the recorded interval between process termination and finalized source/journal evidence. A terminal inspection now awaits the existing completion promise; the initial managed

return uses the same path. It never relaunches a command, fabricates an exit or admits an unknown test report. Four deterministic post-exit controls fail against the sealed F70 runtime and pass after the correction; a running immediate-snapshot control passes in both. Related checks pass 53 cases with one platform skip; affected lint and line budgets pass. Two earlier candidate failures were fixture mistakes: its ordinary command matched a declared test script. Corrected fixtures and original logs are retained. This establishes local publication correctness, not global model success. Evidence: `docs/experiments/light-harness-finalized-process-f71-2026-10-27`.

F72 separately registers six model assignments on the unchanged, known F70 filtering request: two draws each for current Light, Light with bounded incremental inspection waiting, and integrated Cline. F71 is common to both Light arms. The proposed wait ends on new output, termination, or 30 seconds; an omitted cursor permits an immediate snapshot. Existing command timeout, cancellation and SDK loop protection remain intact. Sources, specification and seven grading endpoints must match F70; cache-free images and physical grader copies are reused. The mechanism was first implemented opt-in; current source enables it by default, with an explicit false flag retaining immediate incremental snapshots. Output, timeout, command cancellation and observer-only cancellation controls pass; cancelled observation does not cancel or relabel the owned process. In an actual SDK loopback replay, reference makes six local HTTP requests and aborts at the duplicate-call guard; waiting makes four, launches the command once and transmits exit zero. Initial tools/system hashes and previously forwarded result prefixes match. These are mocked responses, not model trials. The integrated runtime passes 1,207 tests with 13 skipped controls, runner 37/37, affected lint and line budgets.

F72 arm	Source/audit	Tokens	SDK USD	Seconds	Requests
Reference	2/2	1,599,145	.091169	405.5	46
Wait	2/2	1,104,452	.065114	337.6	37
Cline	2/2	2,261,732	.117721	439.3	35

Counts are out of two here: all six official, finite source-endpoint and audited outcomes pass, with all four required invocations observed on final source. Unknown Vitest 5 reporting remains unknown; this does not admit project-wide verification. No failure or missing consumption is dropped. Wait/reference tokens, SDK estimates and duration fall 30.9/28.6/16.7%; request count falls 19.6%. Each repetition has lower tokens (-18.5%, -40.7%). Against Cline, tokens fall 51.2%, estimates 44.7%, duration 23.1%, while requests rise 5.7%; individual token effects are -49.1% and -53.2%. These are two dependent draws on one known issue, not unseen-task confirmation or a general superiority proof. No leave-task-out estimate exists.

Managed inspections fall from seven to four. Reference emits five running snapshots with no raw cursor progress; candidate emits zero. A displayed empty string can still contain raw newline progress: one such candidate observation is retained separately, never labelled fabricated or unchanged. The natural reference draws do not hit the guard; preventing false termination is demonstrated by the fixed SDK replay, not a new live success-rate gain. Other model decisions also vary, so not all nine saved requests or token savings are attributable to polling. Observed tool-time unions grow from 92.2 to 106.9 seconds while HTTP intervals fall from 291.3 to 210.9; this localizes the exchange without separating generation, provider queue, transport or cache causes. Observer overhead is reported without subtraction. Keep the wait as the default after local contract controls and the matched-quality panel; preserve the explicit disable flag and require fresh issues before a broader effectiveness claim. F72 adds estimated USD0.27400372; cumulative observed SDK estimates are USD13.34553584, with USD18.64 historical unknown reservations kept separate from measured spending. Protocol: `docs/experiments/light-harness-inspection-wait-f72-2026-10-27`.

A uniform grader-scope sensitivity is registered after the first F72 checkpoint. The upstream React compiler includes `test-setup.ts`, which imports the Jest DOM augmentation; the isolated delivered-test compiler omits it. All twenty filtering patches reproduce their original compiler

outcomes before the same original-environment type audit. Only F70's first reference changes from TS2339 to exit zero: its source-only sensitivity becomes 4/4 rather than 3/4; its original 2/4 audited count stays fixed. The other F70 authored TS2353 error remains failed. Every F69/F72 classification is unchanged. No model patch is repaired and independent compilation never retroactively proves agent execution.

All 46 unchanged patches reproduce their finite endpoint exits and compiler diagnostics (28 original, twelve F70, six F72), with exact source/permission reconstruction and zero mismatches. The default-on runtime/runner checks pass again. Five additional real Vitest 3.2.4 wrapper controls pass using a private physical dependency copy; shared pinned images are untouched. The standard suite's 13 skips include these five separately exercised cases and eight remaining external-runtime/platform controls.

F73: future grader correction, local evidence only. The input-adapter generator now includes the unchanged upstream React setup when new React tests are selected. Eight old/corrected compiler and selection controls confirm matcher recovery, preservation of a real TS2322 error, unchanged core-only ambient scope, and continued rejection of missing delivered tests. This fixes a grader defect, not an agent source defect. Historical input bytes, model outcomes and required-execution proofs stay frozen; future campaigns require a new input version. Evidence: `docs/experiments/light-harness-react-test-environment-f73-2026-10-27`.

39 Next empirical priorities

Confirm process-observation and efficiency effects on unseen issues and another project, with the original test environment faithfully represented. Then investigate source/test-result crowding and adoption of grouped reads, which remain unused in the twenty F68–69 Light draws. Strengthen negative type-contract coverage and keep execution, passing, formal admission and user acceptance separate. Distinguish model decisions from unavailable information and contract-selection friction. Keep success primary, isolate changes, retain failures and unknown consumption, and test omission sensitivity before general efficiency claims. Optional Z3 witness translation/replay stays independent. No finite panel establishes that the remaining improvement margin is exhausted.